

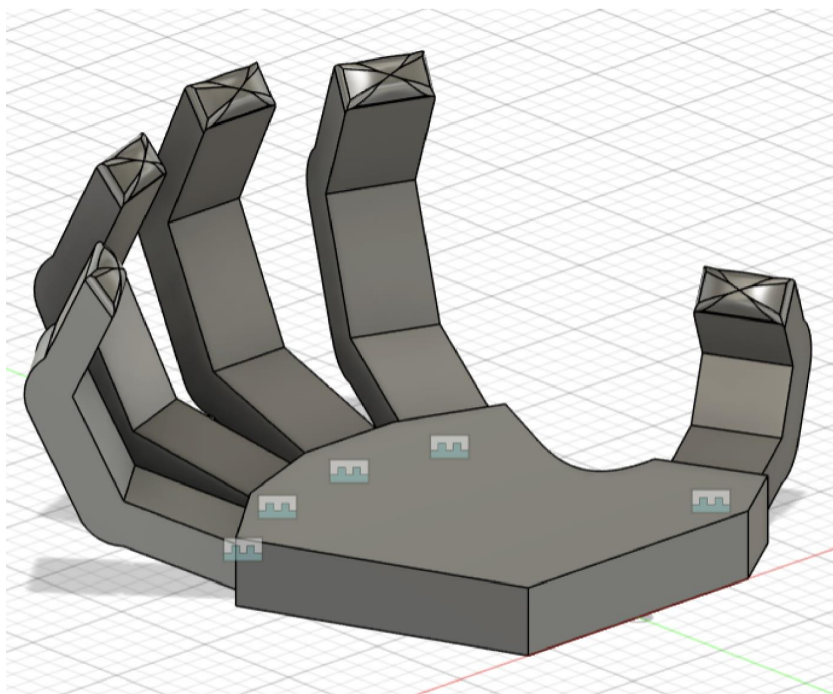
CAD Workshop: *Autodesk Fusion*

Project Overview

Welcome to the Purdue MIND CAD Workshop! In this project, you will be designing a **prosthetic hand** using Autodesk Fusion 360. Rather than simply recreating geometry, you'll learn how engineers create parametric, scalable, and fully constrained models that can be easily modified throughout the design process.

The skills introduced through this workshop are foundational for many of the projects our subteams will be pursuing this year and reflect modeling practices used in professional product development. Autodesk Fusion is particularly great for beginners, as it allows you to easily share designs and collaborate on files with your team, and is available on both Windows and macOS.

Estimated time: 2-3 hours



Fusion Installation

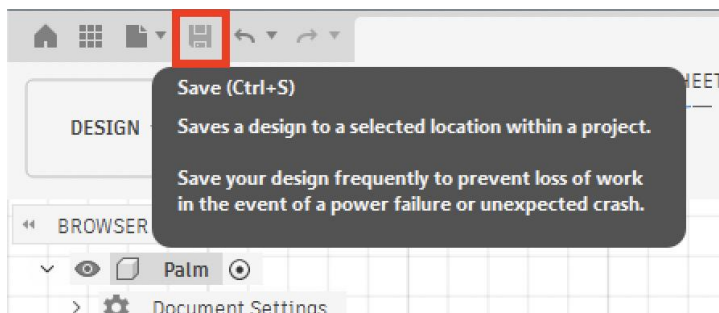
- 1) Navigate to
<https://service.purdue.edu/TDClient/32/Purdue/KB/Article/2276/Autodesk-Fusion-360>
- 2) Follow the instructions to create an Autodesk account.
- 3) At the bottom of the page under “Personally-Owned Systems”, there are two links to both the admin installer and the commercial installer. Use either one to begin installing the software.
 - a. If using the commercial installer, the link will take you to an Autodesk webpage. Click the second option on the webpage “Fusion”.
- 4) You can also access Fusion through your browser using this link:
<https://fusion.online.autodesk.com/>
- 5) Sign into Fusion using the Autodesk account you created

General Tools

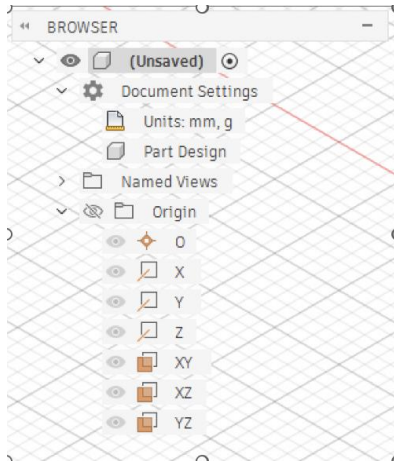
Saving

Fusion does not automatically save your progress. Make sure to save your design as frequently as possible to avoid losing your work!

- **Keyboard Shortcut:** Ctrl-s



Browser Menu



- The Browser shows and organizes the different parts of your design.
 - As you create **sketches, bodies, and components**, they will appear here.
 - Click the **eye icon** next to an item to **show or hide it**.
 - Use the arrows next to folders to **expand or collapse** them.
 - The **Origin** folder contains the X, Y, and Z axes and planes that help you position and create your model.

Bottom Menu



- Orbit
 - Allows you to **rotate your view around the model in 3D space**, so you can view it from different angles and sides.



- Pan

- Moves your view **side to side or up and down** without rotating the 3D model, allowing you to reposition your design on the screen.



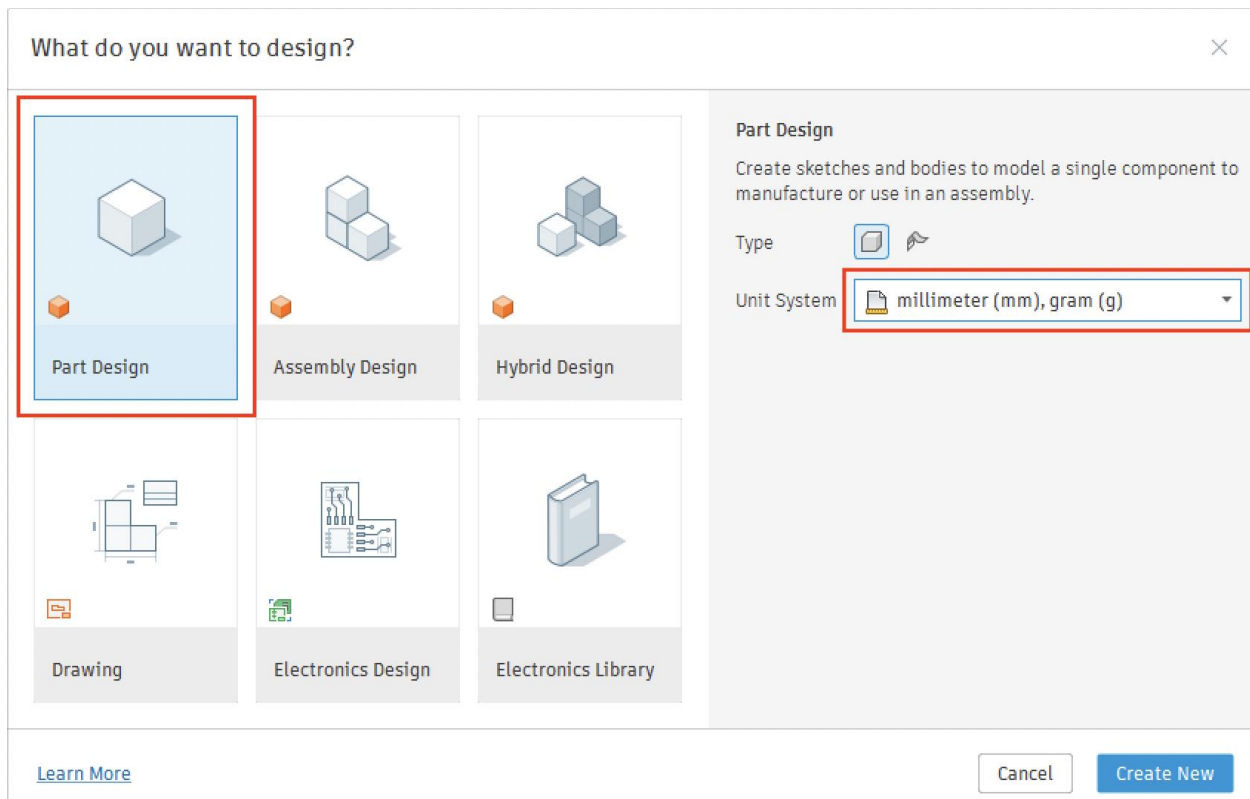
Part 1: *Creating Parameters*

1. Open **Fusion** on your computer
2. Under “**Start Working**” select “**New Design**”

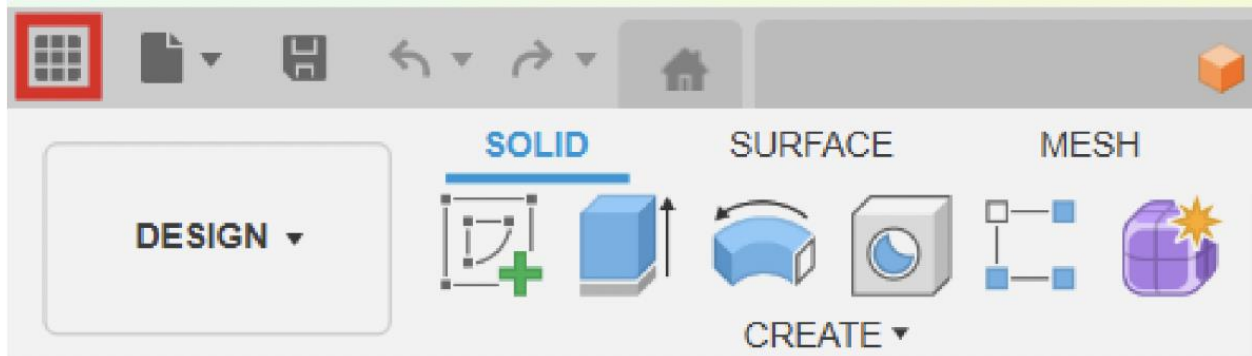
Start working



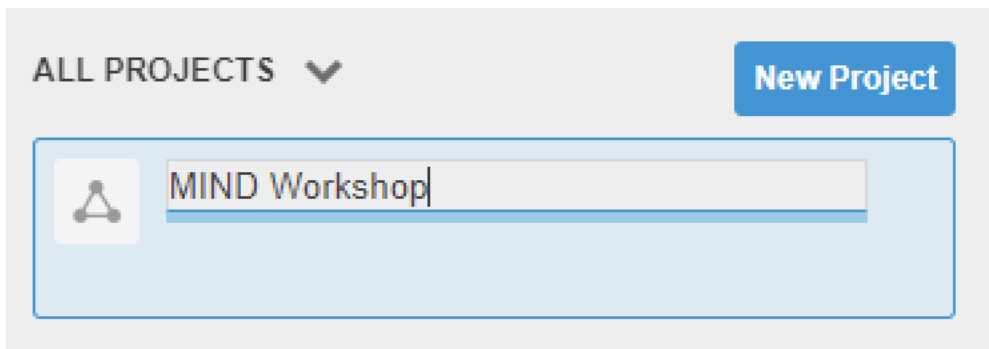
3. Select “**Part Design**” and ensure your unit system is in millimeters and grams.
 - a. This should open a blank, untitled workspace.



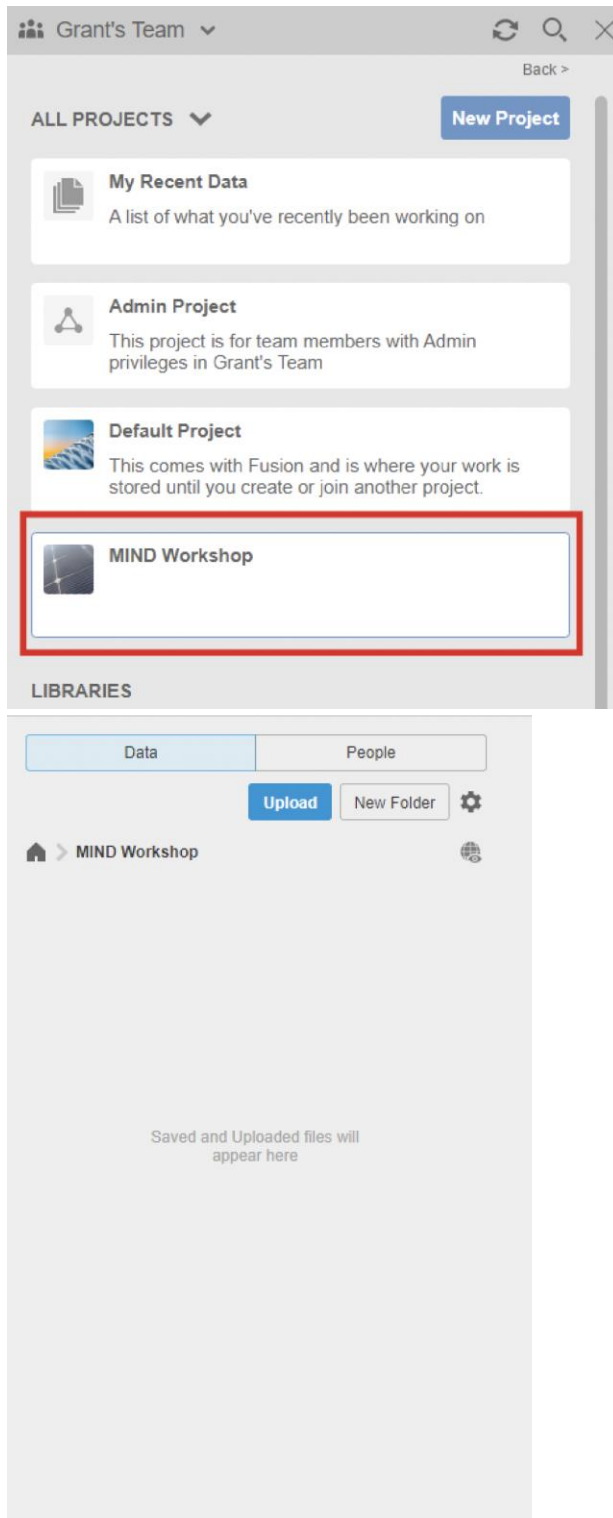
4. Find the **3x3 dot matrix** icon in the upper left corner.



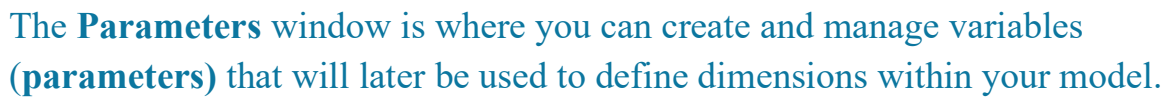
5. This will open your data panel. Select the blue “**New Project**” button, title your new project “**MIND Workshop**”, and press enter.



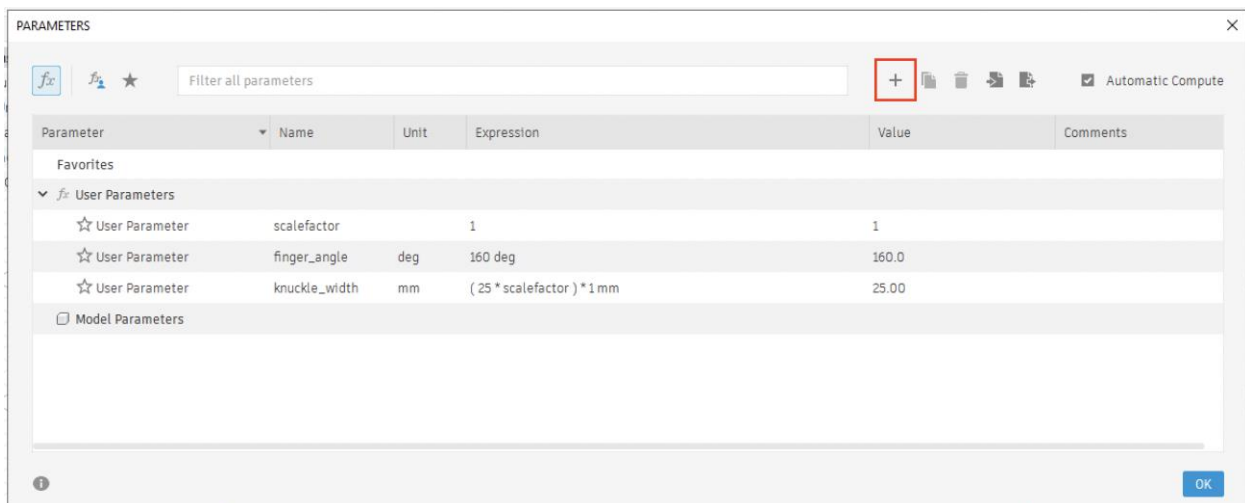
6. After your new project is created, it will automatically pop up in your project list. Now double-click on your new project to open its data space.



8. Navigate to the “**Modify**” menu at the top and select “**Change Parameters**”.



9. Click the "+" **button** and make parameters for the following quantities
- scalefactor: 1 (unitless)
 - Controls the overall scale of the palm. Because it is unitless, it acts as a multiplier. A value of 1 represents the original size of the model.
 - finger_angle: 160 degrees
 - Defines the angle used to create the finger attachment geometry. Instead of repeatedly entering 160°, you will reference finger_angle wherever this value is needed.
 - knuckle_width: 25mm * scalefactor
 - Defines the width of each knuckle section. It includes scalefactor so the knuckle dimensions automatically resize with the rest of the palm.



Add User Parameter

Name: scalefactor

Unit: No Units

Expression: 1

Value: 1

Comment: Enter Comment

OK Cancel

Add User Parameter

Name: finger_angle

Unit: deg

Expression: 160

Value: 160.0

Comment: Enter Comment

OK Cancel

Add User Parameter

Name: knuckle_width

Unit: mm

Expression: 25 * scalefactor

Value: 25.00

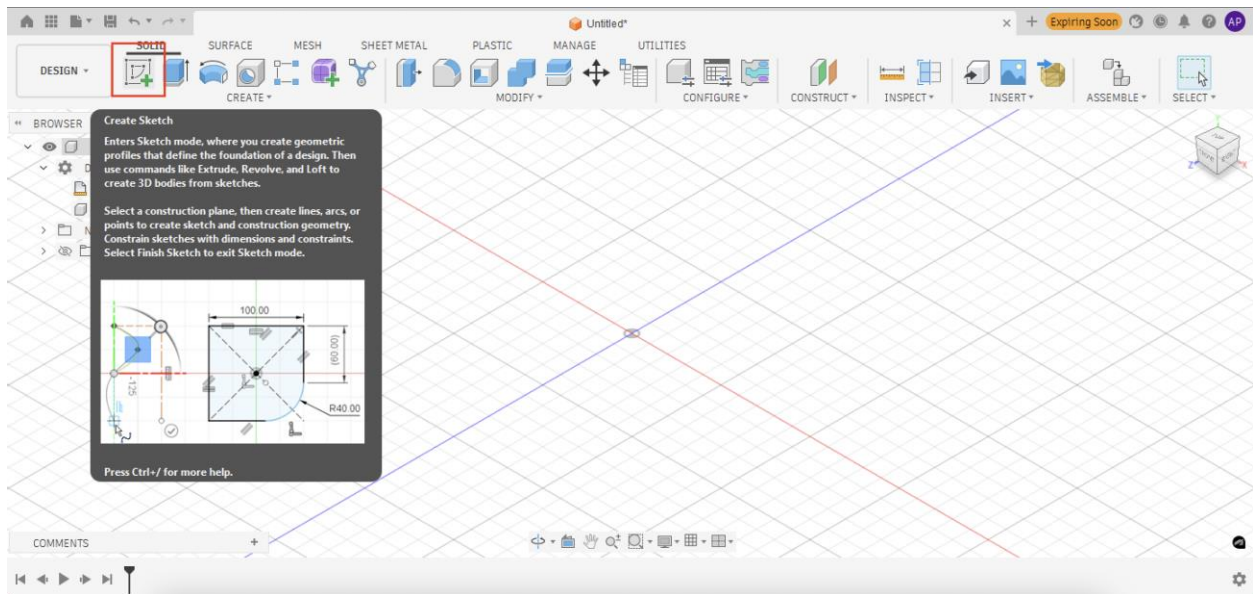
Comment: Enter Comment

OK Cancel

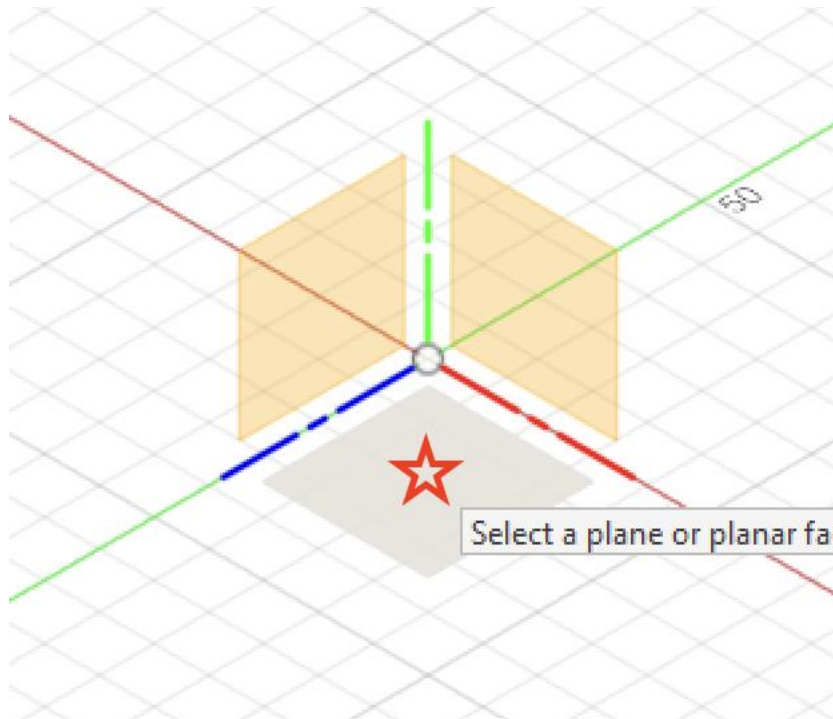
10. Once all parameters have been made, exit the menu by pressing "OK"

Part 2: *Creating the Palm Sketch*

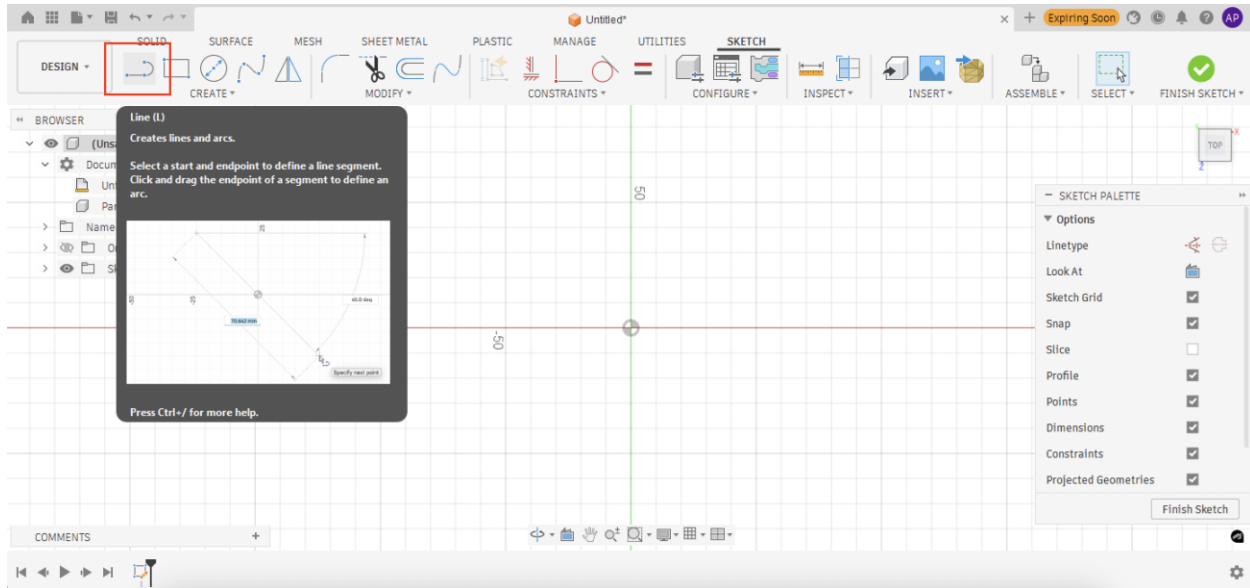
11. Click on the **"Create Sketch"** button (first option in the top menu)



12. Click the bottom (XY) plane



13. Click on the “**Line**” button (first option in new sketch menu)



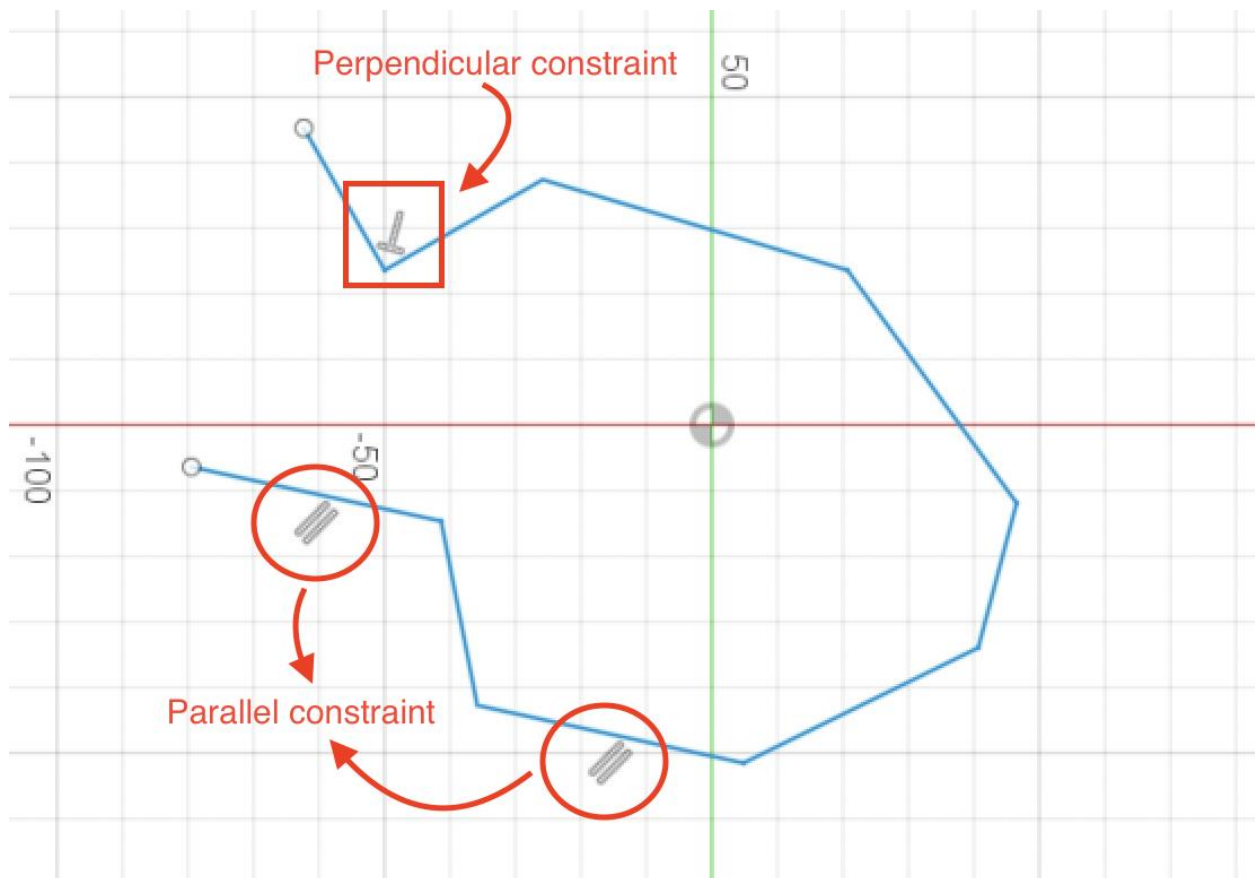
14. Draw 9 connected line segments in **no particular form**.

- When drawing each line, Fusion will generate icons to show you when your segments are perpendicular or parallel to each other. **Do your best to avoid this and randomize your angles.**
- This will prevent Fusion from predicting constraints on your sketch.
- If Fusion does predict constraints, it could make the next step over-constrain the sketch, breaking the model

KEY CONCEPT:

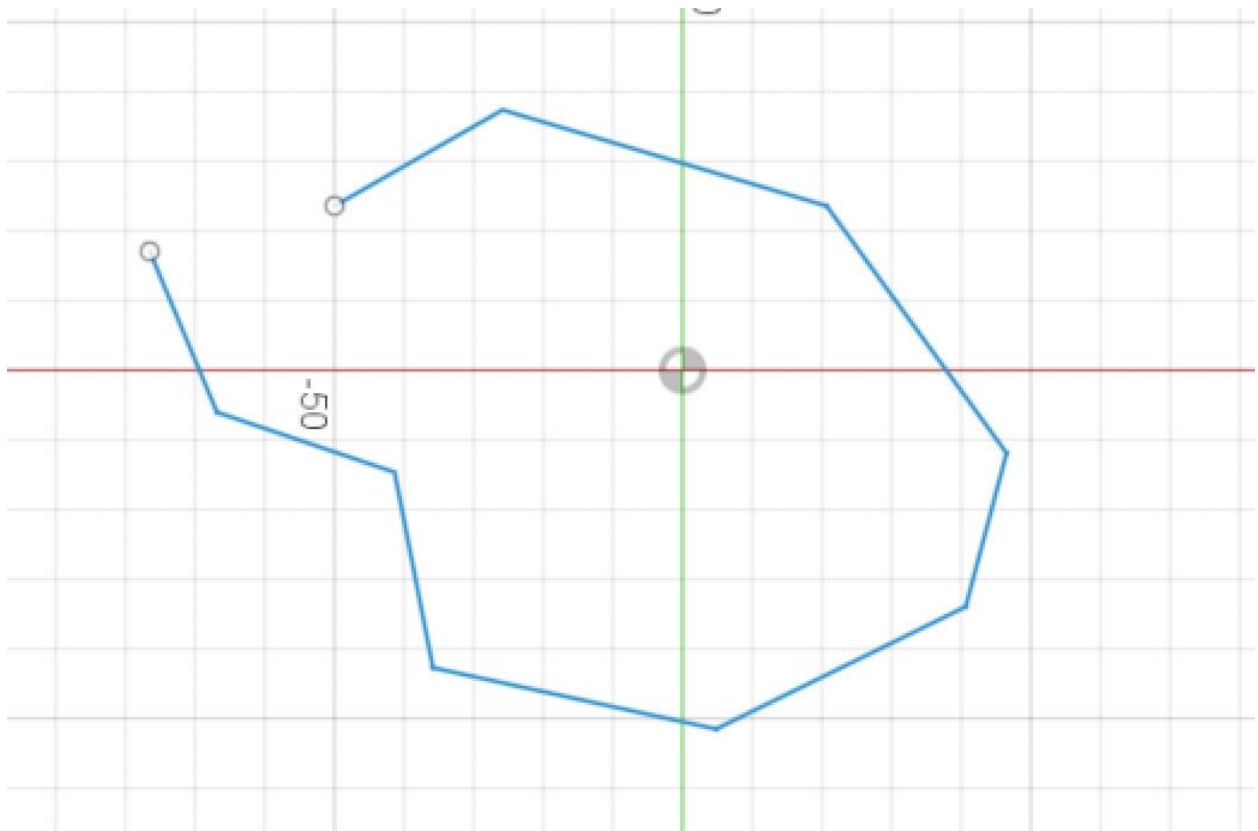
- Geometric constraints** define how sketch elements relate to one another, such as **parallel, perpendicular, horizontal/vertical, tangent, or coincident**. These relationships are maintained even if the dimensions of your design change.
- Dimensional constraints** define the **specific size or position** of sketch elements using numerical values, such as a line being **50 mm long** or two points being **20 mm apart**.

Constrained example:

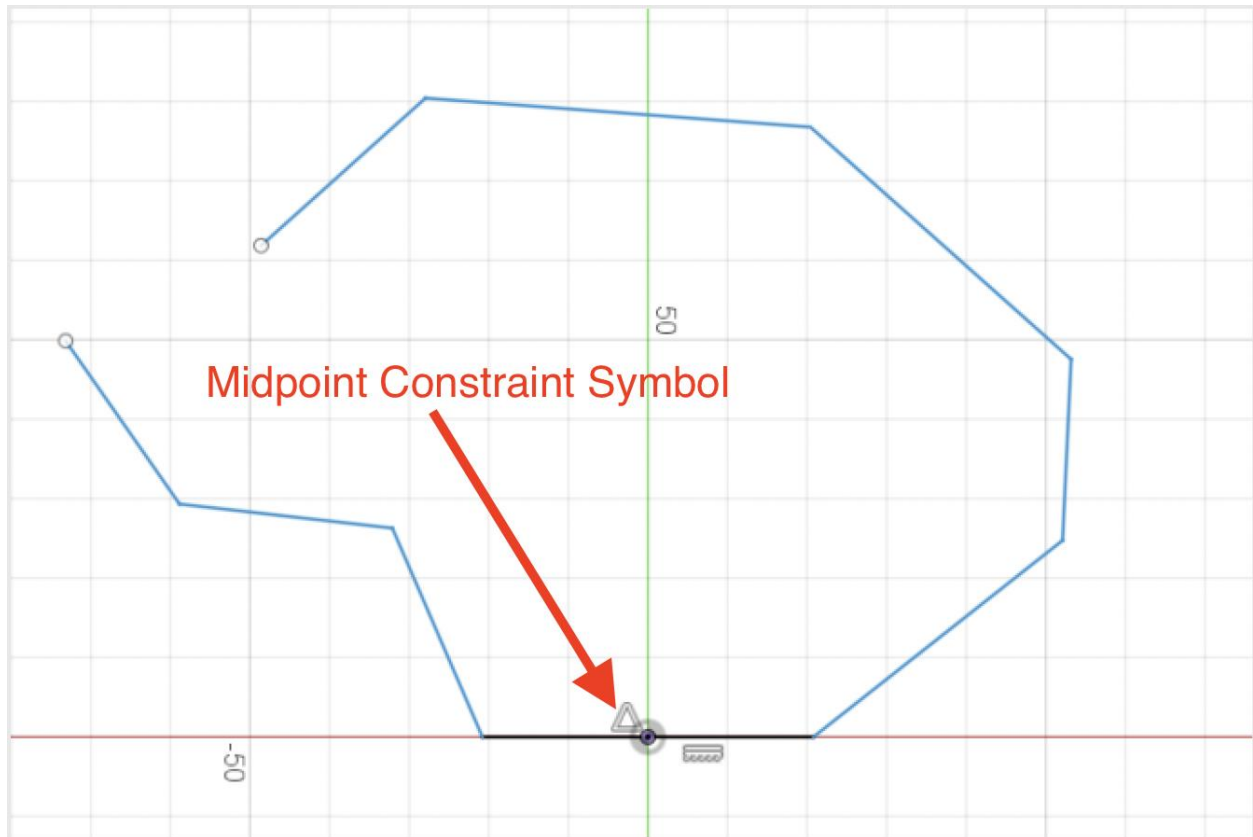


15. If a constraint is autogenerated, a constraint icon is generated (shown in image above). If this happens, simply select the icon and hit the delete button on your keyboard.

Unconstrained example (This is what you want! But it is just a general depiction, so your lines can look like anything!):



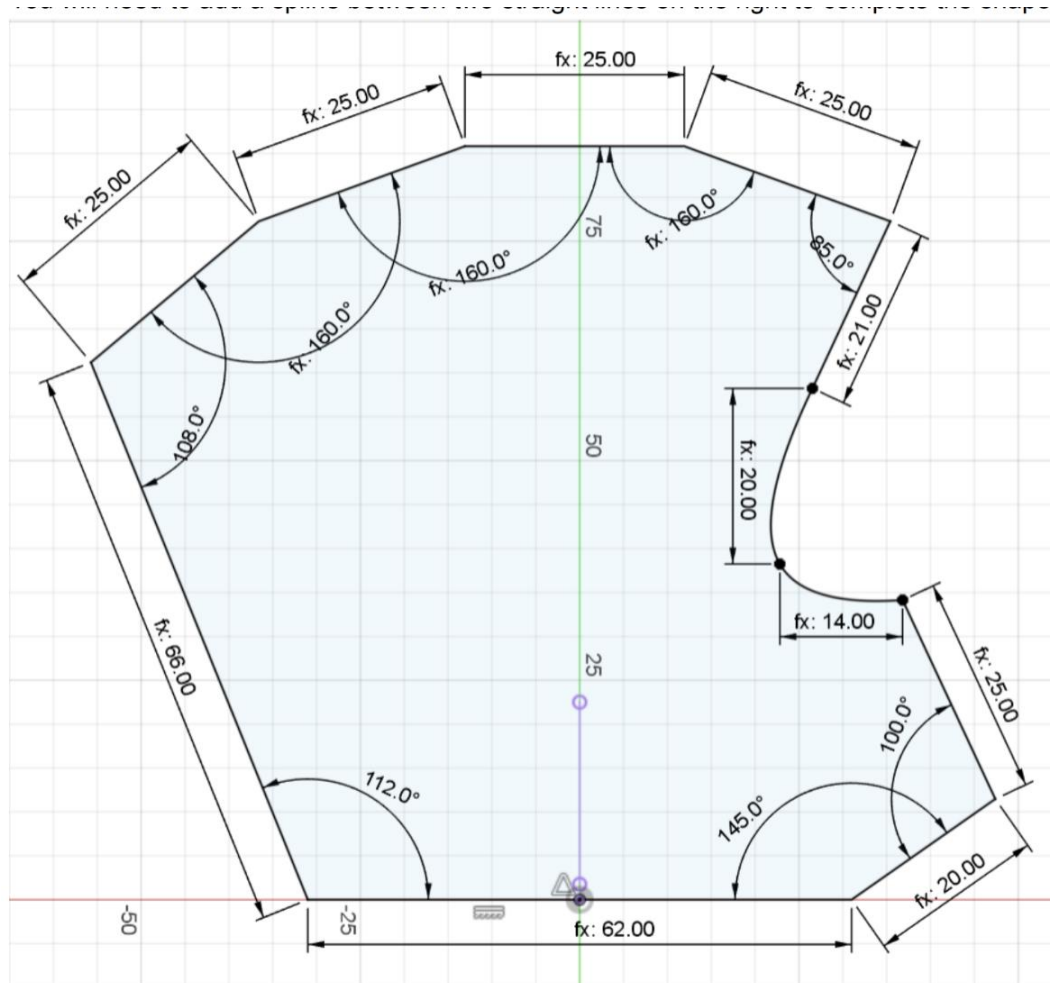
16. Open the “**Constraints**” menu.
 - a. Click on your bottom line and select the “Horizontal/Vertical” constraint. This makes the bottom line parallel to the x-axis.
 - b. Click on the “Midpoint” constraint. Click on your bottom line (Fusion will highlight it) and then click on the origin of the graph. This centers the sketch on the x-axis



17. Open the **“Create”** menu at the top of your screen. Select **“Sketch Dimension”**. This allows you to select any line/angle and change its length/angle.
- a. To change length of line segment:
 - i. Select the **“Sketch Dimension”** button and then click on the line you want to alter. Drag away from your clicking point, and click to place your dimension value. The length will automatically highlight, so you can simply type in your value and press enter.
 - b. To change degree value of angle:
 - i. Select the **“Sketch Dimension”** button and then click on the two lines that create the angle you want to alter. Drag away from your clicking point, and click to place your dimension value. The degree value will automatically highlight, and you can simply type in your value and press enter.

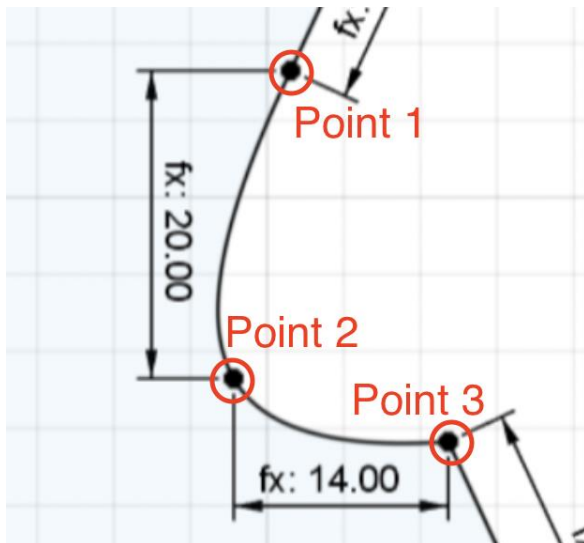
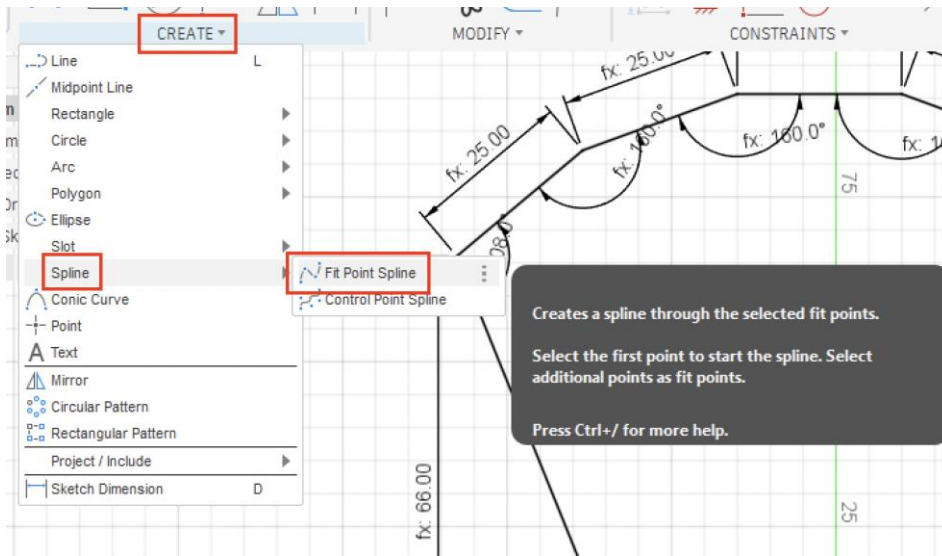
18. Match the following dimensions shown in the image below:

- All measurements that are 25mm should use the `knuckle_width` variable
- All 160 degree measurements can use the `finger_angle` variable
- All dimensions marked with "fx:" should be multiplied by `scalefactor` (other than those made with `knuckle_width`)
- We will create the curved portion that completes the shape in the next part!



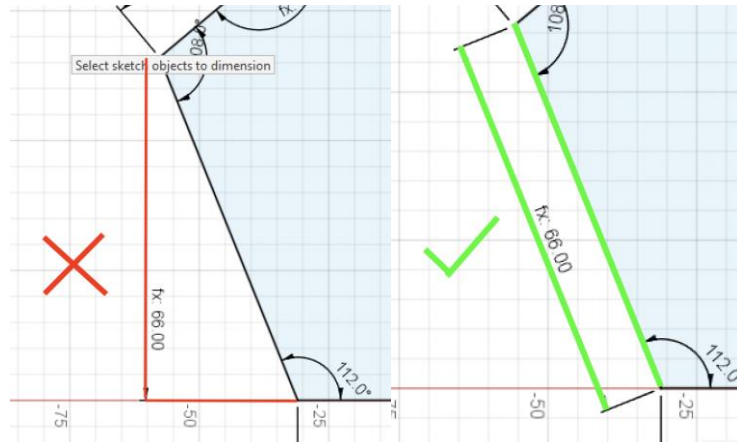
19. To connect the shape (curved lines shown above), create a spline. Navigate to the **“Create”** menu and hover over the **“Spline”** option. Find the **“Fix Point Spline”** option that pops up in the side menu and click on it.

- Click to place your 3 points as shown in the image below. You can roughly estimate the location of Point 2 when initially drawing your spline.
- Use the **“Sketch Dimension”** tool along with your **scalefactor** variable to match the dimensions shown in the image.

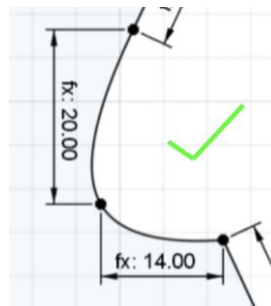


CHECKPOINT:

- Ensure all the length dimensions you inserted are parallel to the respective line segment. Incorrect dimension placement can measure the **horizontal or vertical distance between two points** rather than the **actual length of the line segment**, resulting in incorrect palm geometry.
 - Ex:



- BUT, your dimensions for the **spline** should measure vertical/horizontal distance.

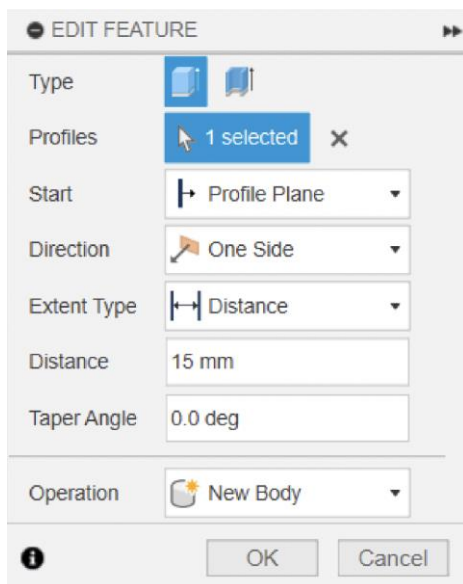


- Compare your dimension placement to the reference sketch before continuing.
- When you are finished, you should rename the sketch based on its purpose. In this case, we will call it “Palm”.
 - Right-click on the current sketch name (should be “Sketch1”) to open a menu. Find the “Rename” option and alter the text. [You can use this menu in the future to edit your sketch, rename it, etc.](#)

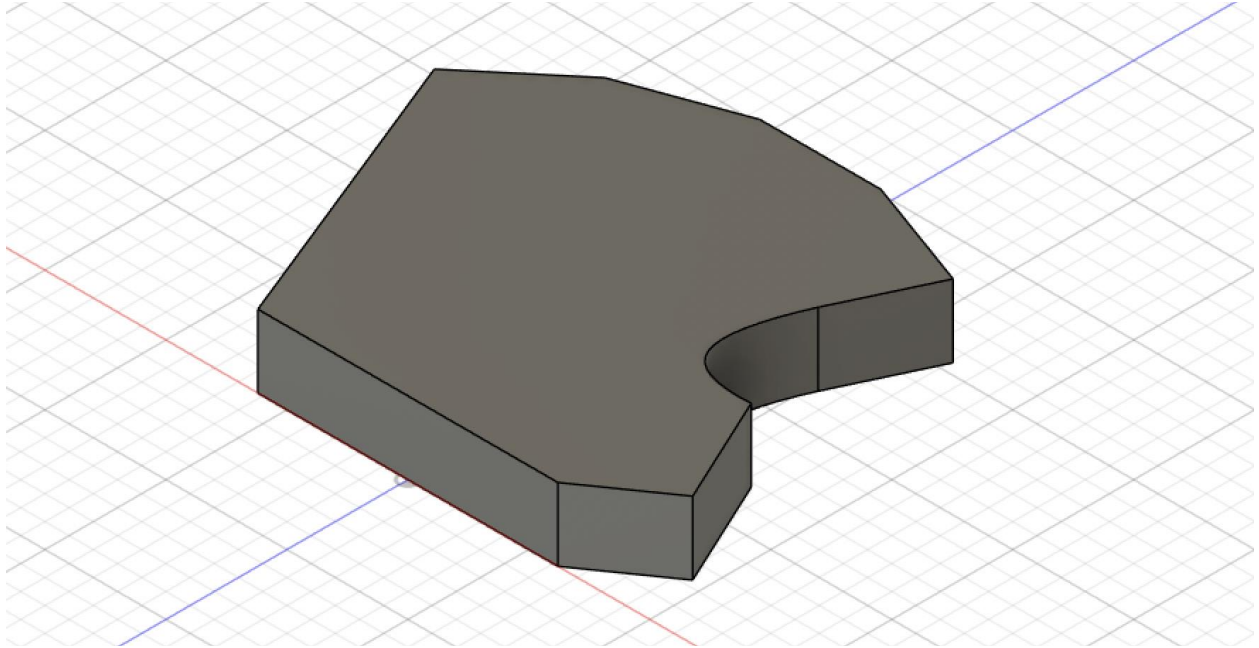
20. Click on the “**Extrude Tool**”



21. Enter "15mm" for the distance, everything else should default to the correct values. Click “OK” after you finish.



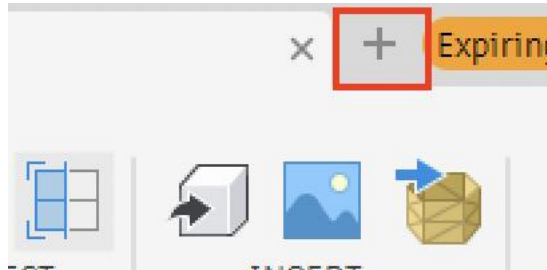
Extruding adds depth to a 2D sketch, transforming the palm profile into a 3D solid. The original sketch remains part of the model history, allowing changes to the sketch or its parameters to update the 3D palm automatically.



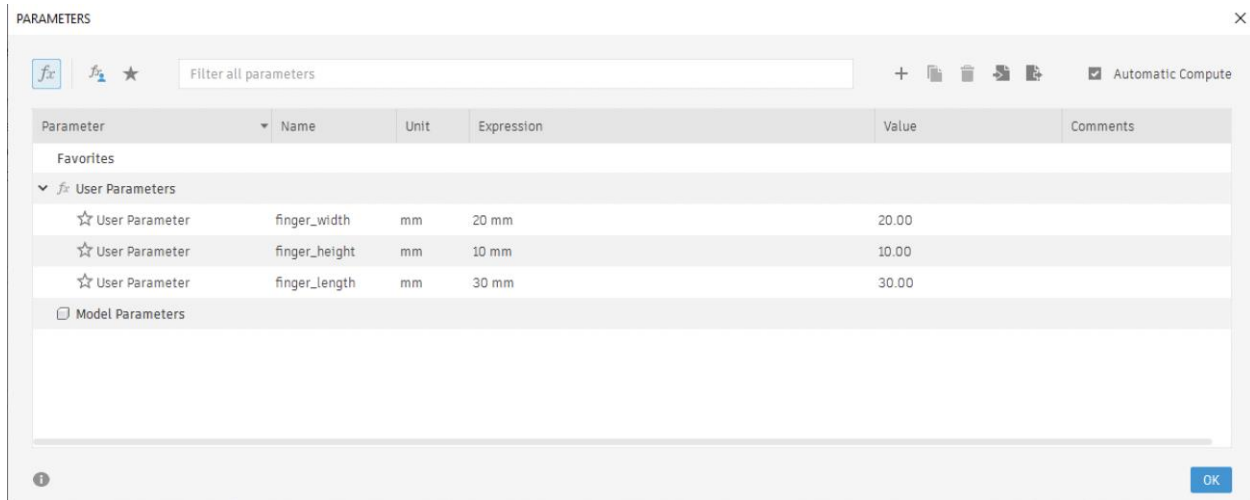
22. **SAVE your current model (CTRL-S)!** Name the design “Palm”.
- This should automatically save your design in the new “MIND Workshop” project you created. Open your data panel (3x3 dot matrix) and click on the project. Your design should now be located in the data space for your project.

Part 3: *Finger Design*

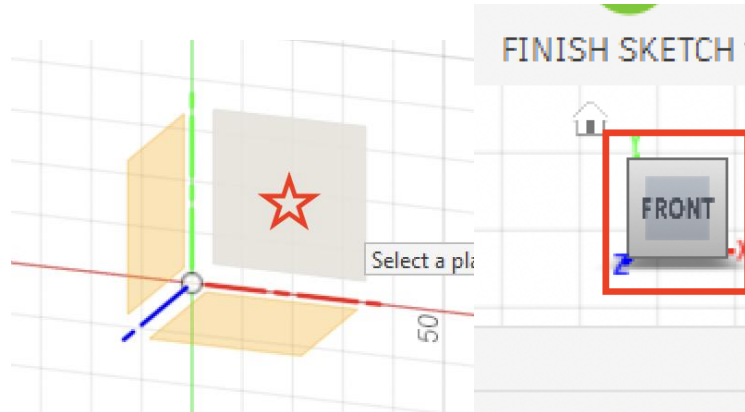
23. Create a new part design by adding a tab in Fusion, near the top of the screen, and selecting “**Part Design**”. This will act as the finger design.



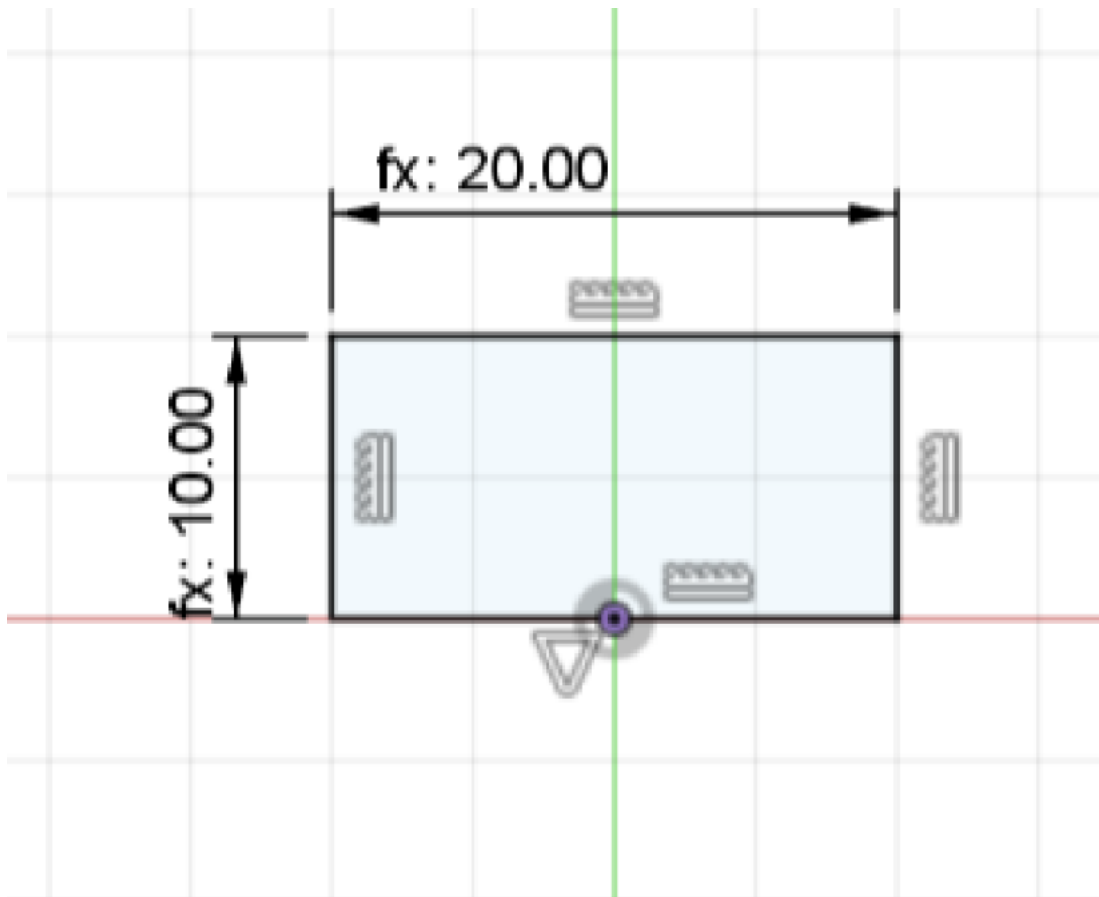
24. Add new variables to this design using the same process as before (described earlier in steps 8 and 9 of Part 1.)
- finger_width: 20 mm
 - finger_height: 10 mm
 - finger_length: 30 mm



25. Create a sketch in the "front" or XZ plane.

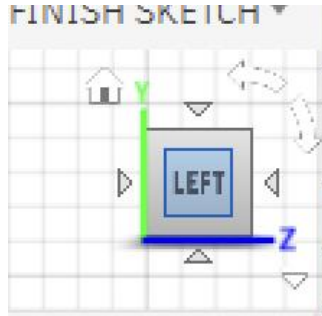


26. Use the **rectangle tool** (option next to the line tool), the **midpoint constraint**, and the **variables defined in step 23** to obtain the sketch below:
- Ensure the 20 mm measurement uses the `finger_width` variable and the 10 mm measurement uses the `finger_height` variable

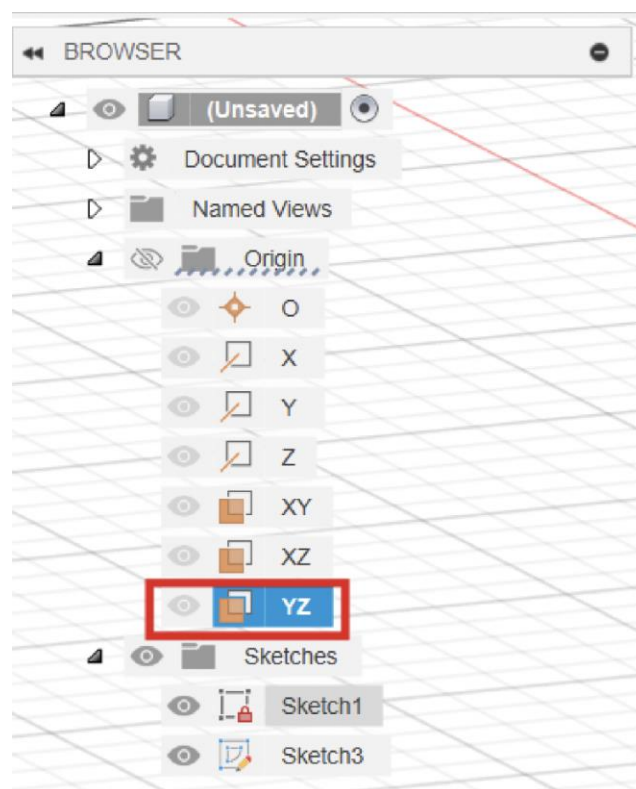


27. Confirm the sketch by pressing the **"Finish Sketch"** button

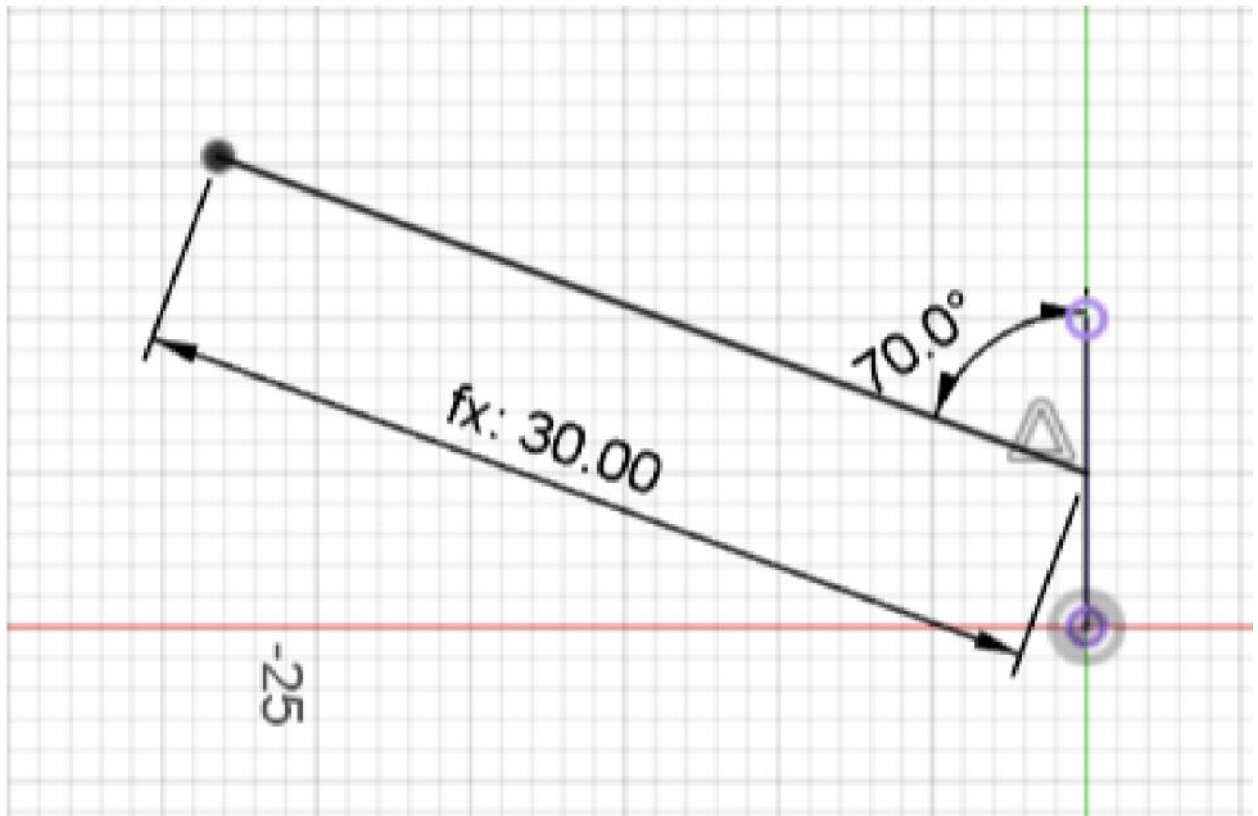
28. Create a new sketch in the YZ plane. You should see the logo below in the top right corner.



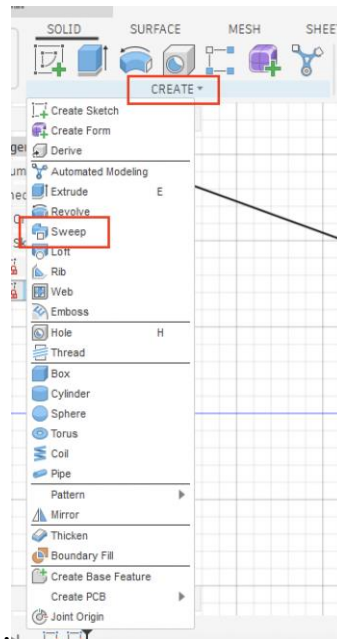
You can also use the menu under “Origin” on the left of your screen to select different planes:



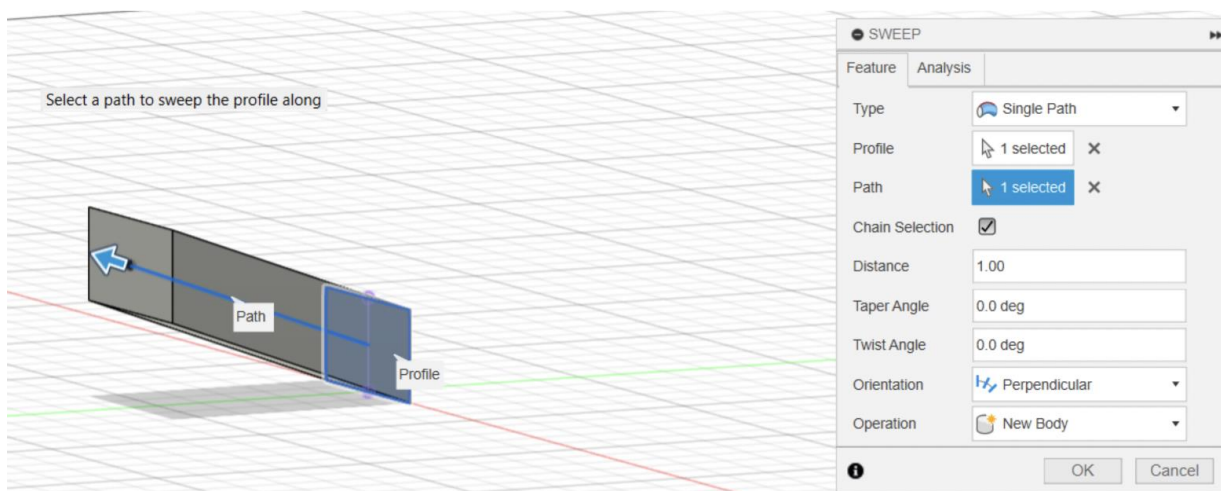
29. In this new sketch, you will see the side of the rectangle you created in the previous sketch. It should appear as a vertical line on the y-axis. Create a line constrained to the midpoint of this line following the below dimensions.
- Remember to use the `finger_length` variable.
 - “Finish Sketch” after you are done



30. Under the “**Create**” section at the top left of the screen, select “**Sweep**”

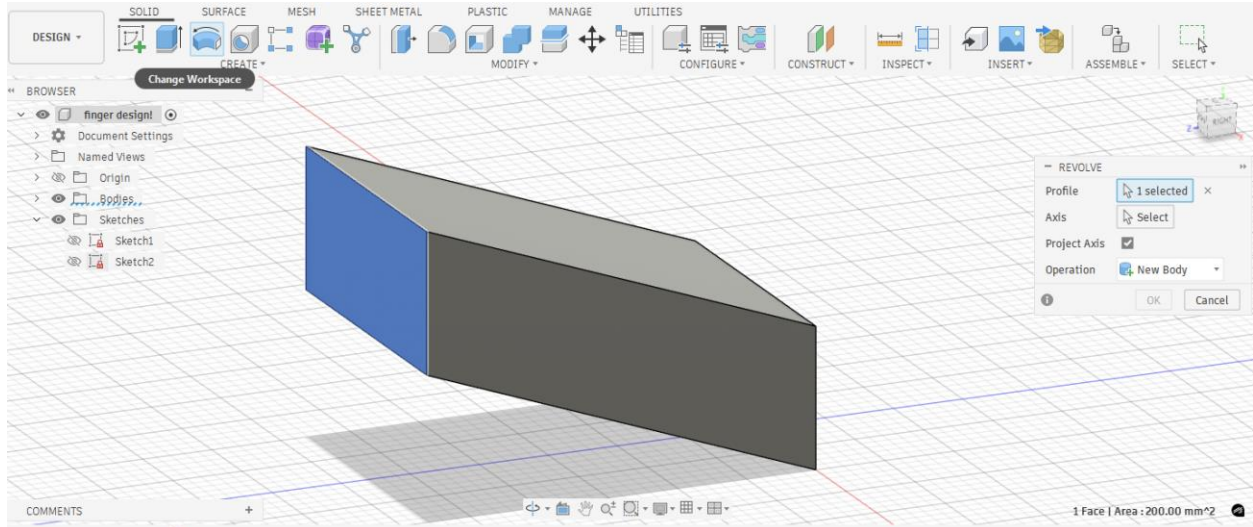


31. Once sweep is selected, choose the face of the original rectangle as the profile and the line from the second sketch as the path.
- Sweeping extends a 2D profile along a defined path. Here, the rectangle determines the **shape and size** of the finger, while the line determines its **length and direction**.
 - Click “OK” after you finish

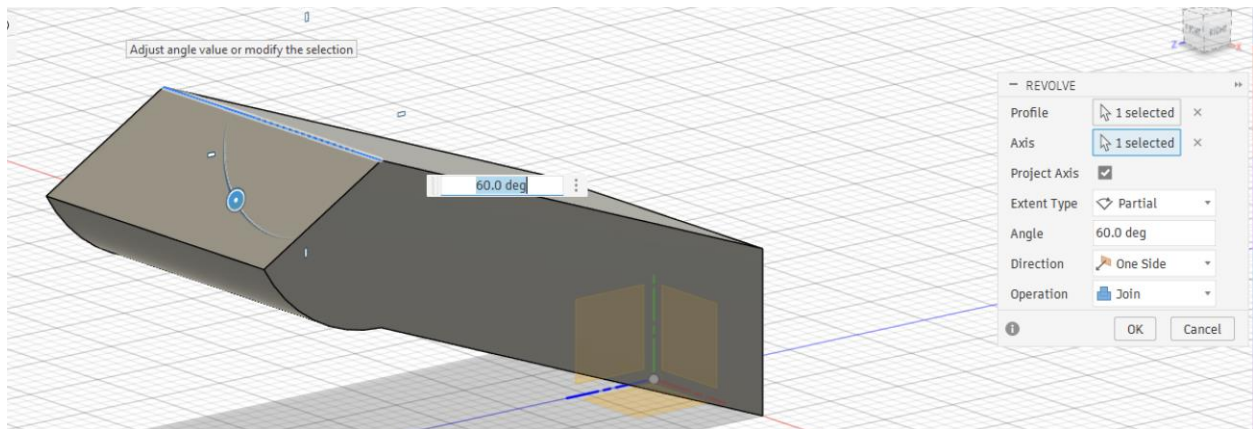


31. Head back to the “Create” menu and now select “Revolve”.

- c. For the “Profile”, select the face of the rectangle created from the previous sweep



- d. For the “Axis”, select the top line of the rectangular face you just selected for the “Profile”

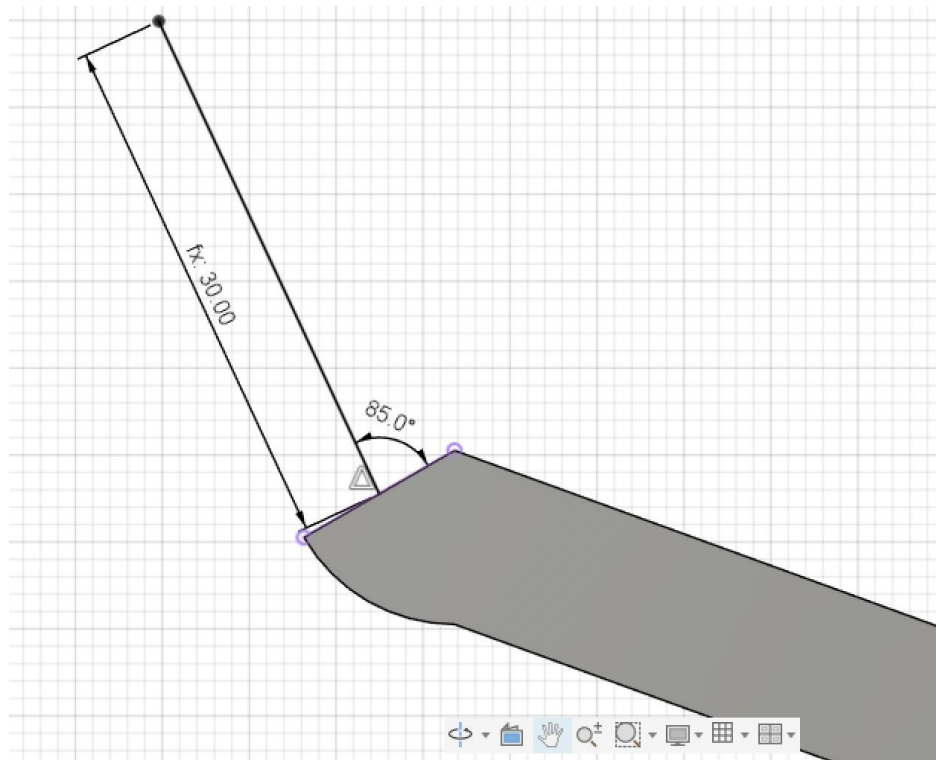


- e. The angle may need to be adjusted from 360 degrees to 60 degrees. Confirm the revolve once the model looks like the above image.

Unlike a Sweep, which extends geometry along a path, a Revolve rotates a profile around an axis. Here, the Revolve creates the **curved transition between finger segments**, giving the finger its bent shape.

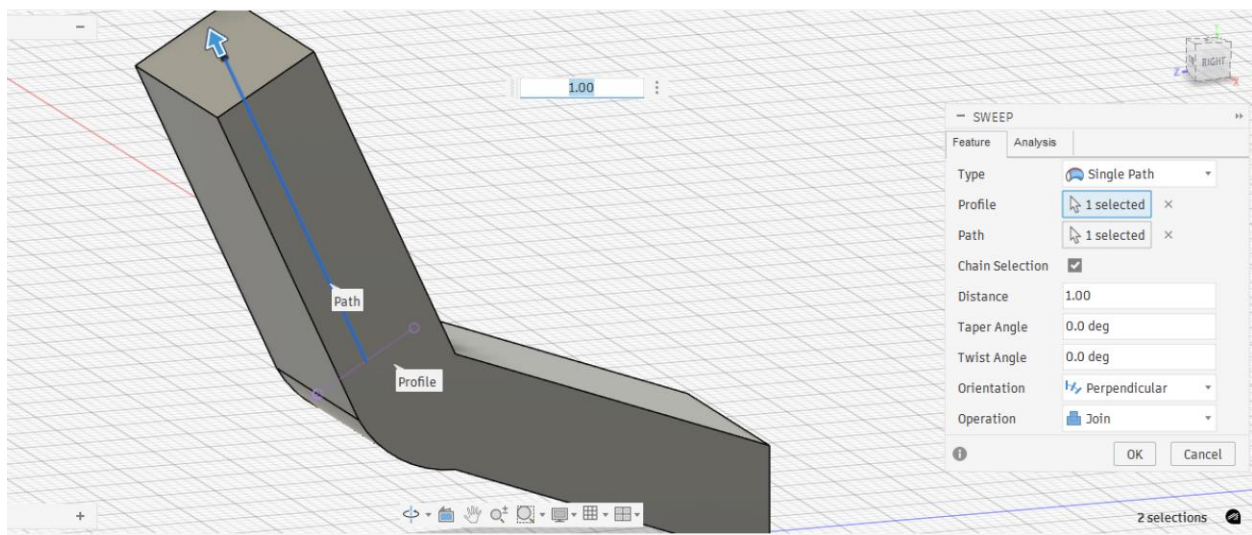
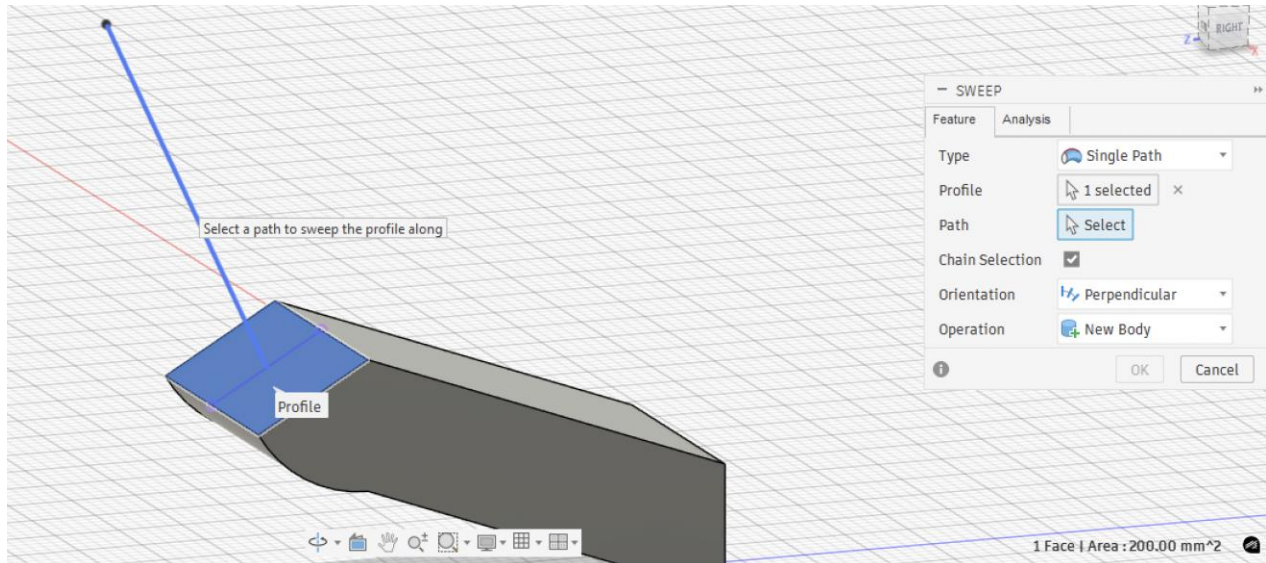
32. Create a new sketch on the YZ plane matching the specifications in the image below

- Remember to use the **finger_length** variable
- Recognize the midpoint constraints in the image!



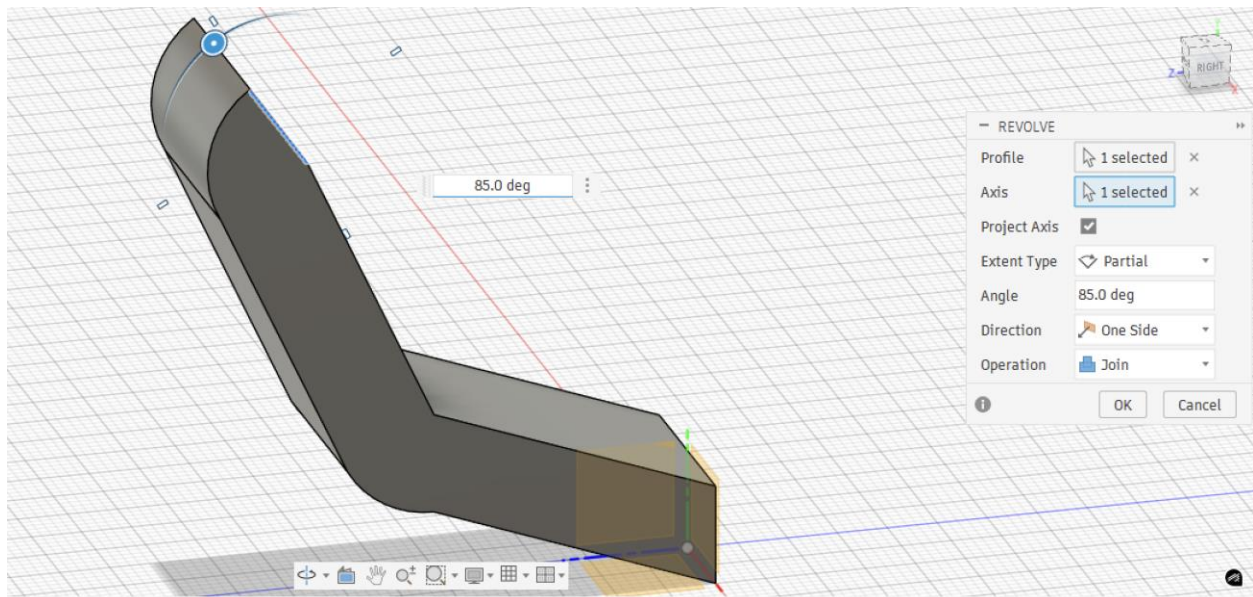
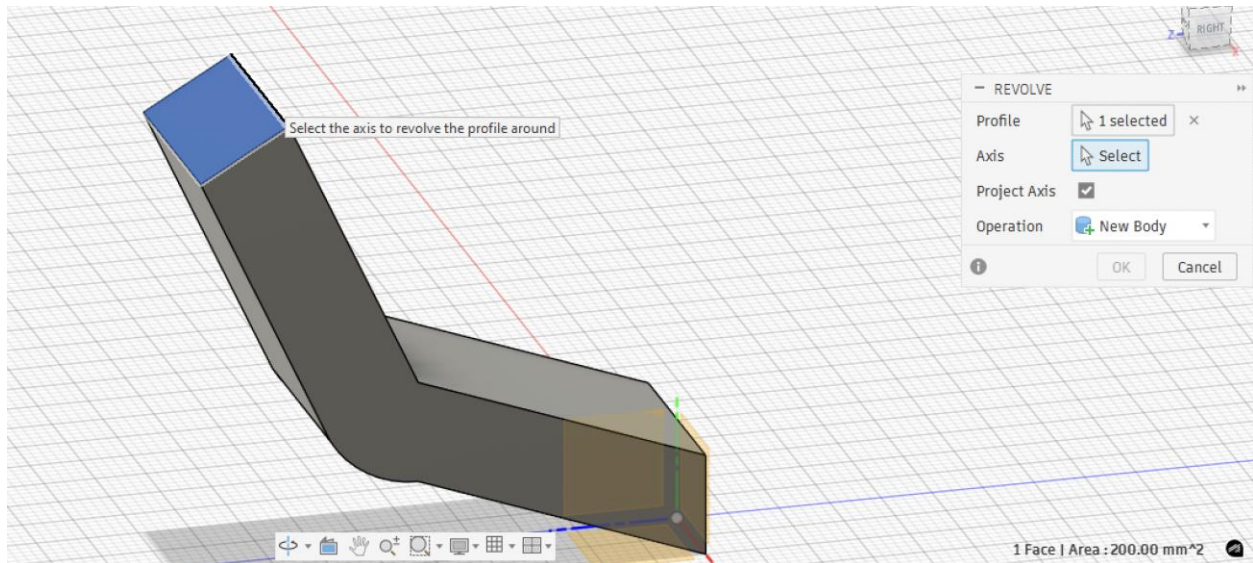
33. Perform another sweep

- a. The profile will be the rectangular plane shown in the image below (face of the revolve you created in step 30)
- b. The path will be the line you just created in the previous step

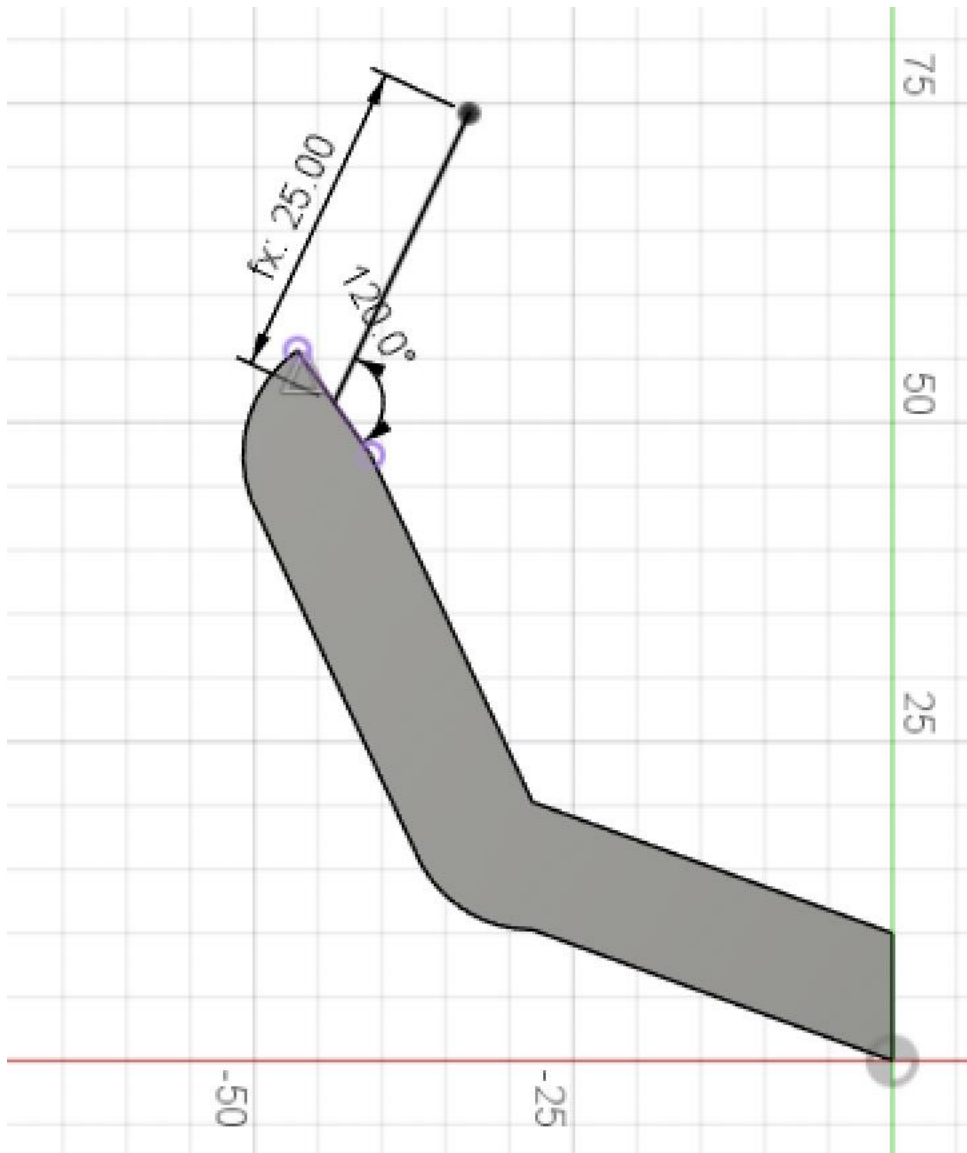


34. Perform another revolve

- Your profile will be the face of the sweep you just created in the last step
- Your axis will be the top line of the same face
- This angle should be 85.0 degrees.

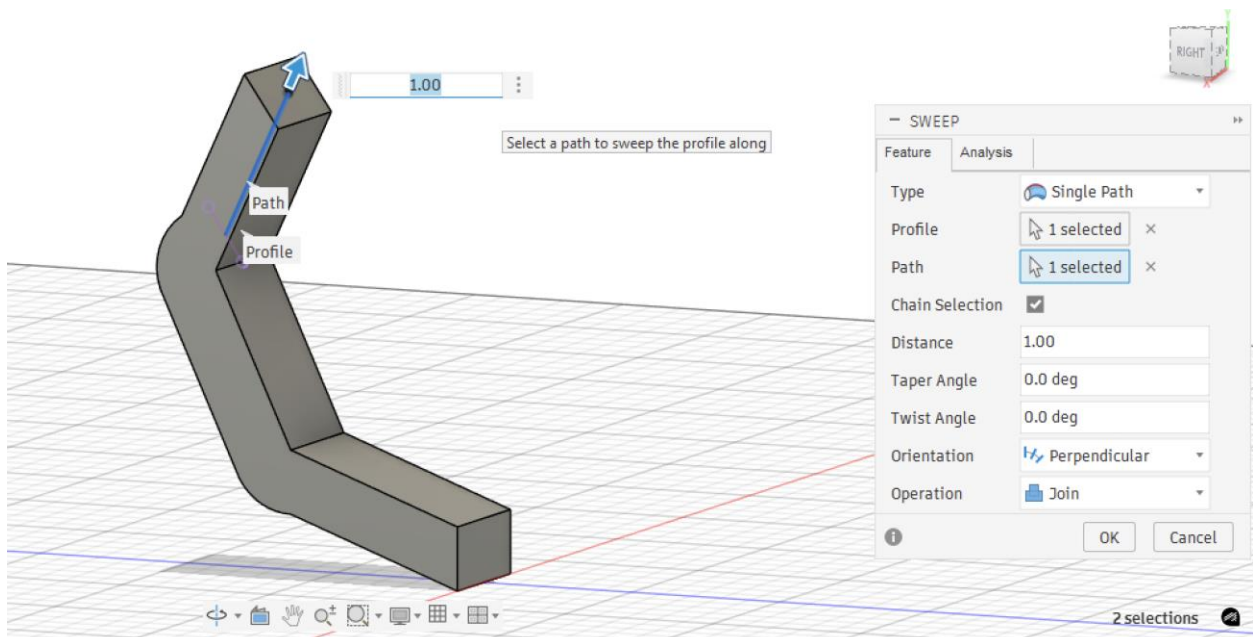
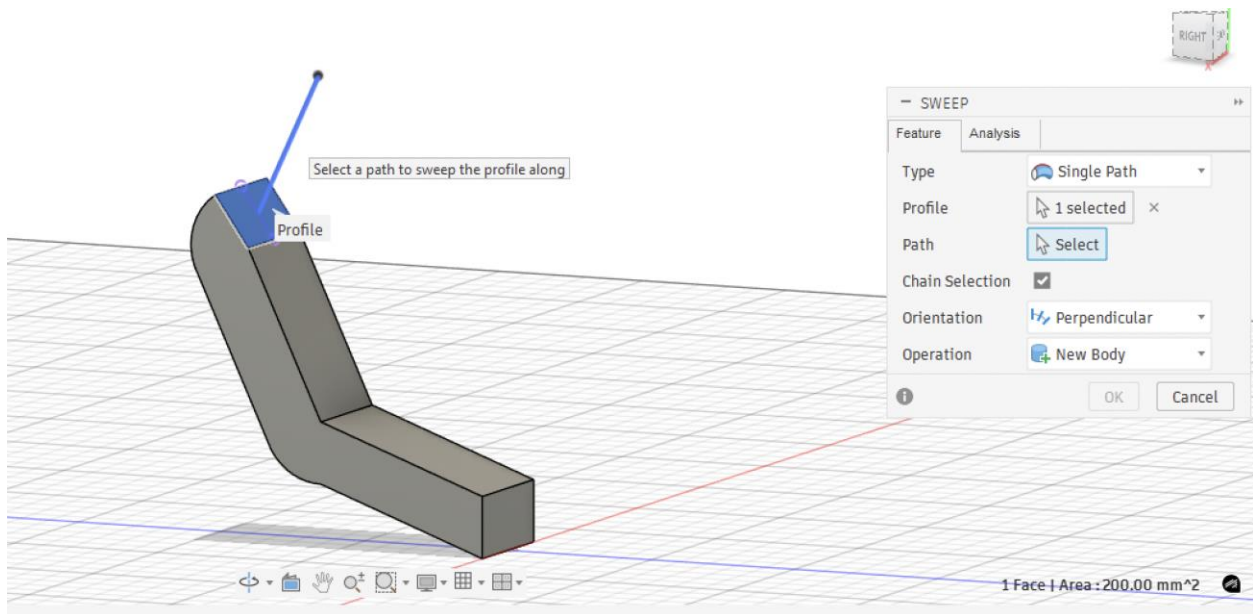


35. Create a new sketch on the YZ plane matching the specifications in the image below.
- Try to use the **finger_length** variable to obtain the specified length.
 - Hint: What can you multiply finger_length by that will result in the specified number?
 - Recognize the midpoint constraints in the image!**

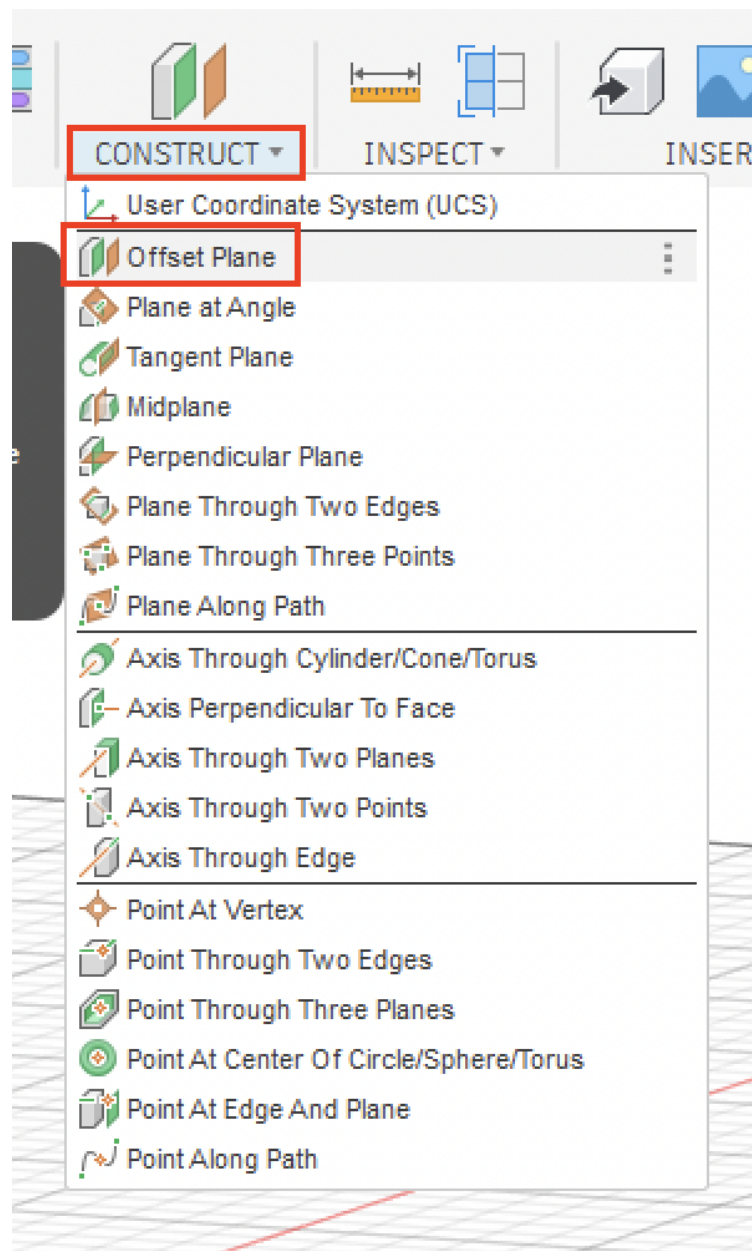


36. Perform another sweep

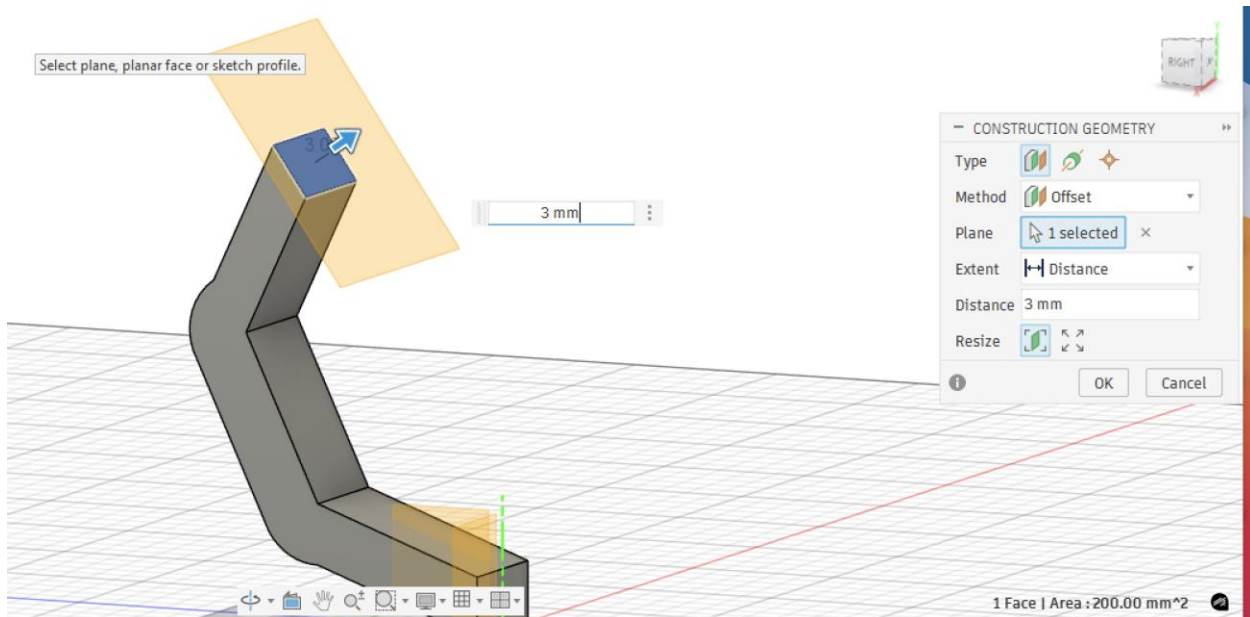
- a. The profile will be the rectangular plane shown in the image below
- b. The path will be the line you just created in the previous step



37. Navigate to the “**Construct**” menu at the top of your screen and click on the “**Offset Plane**” option

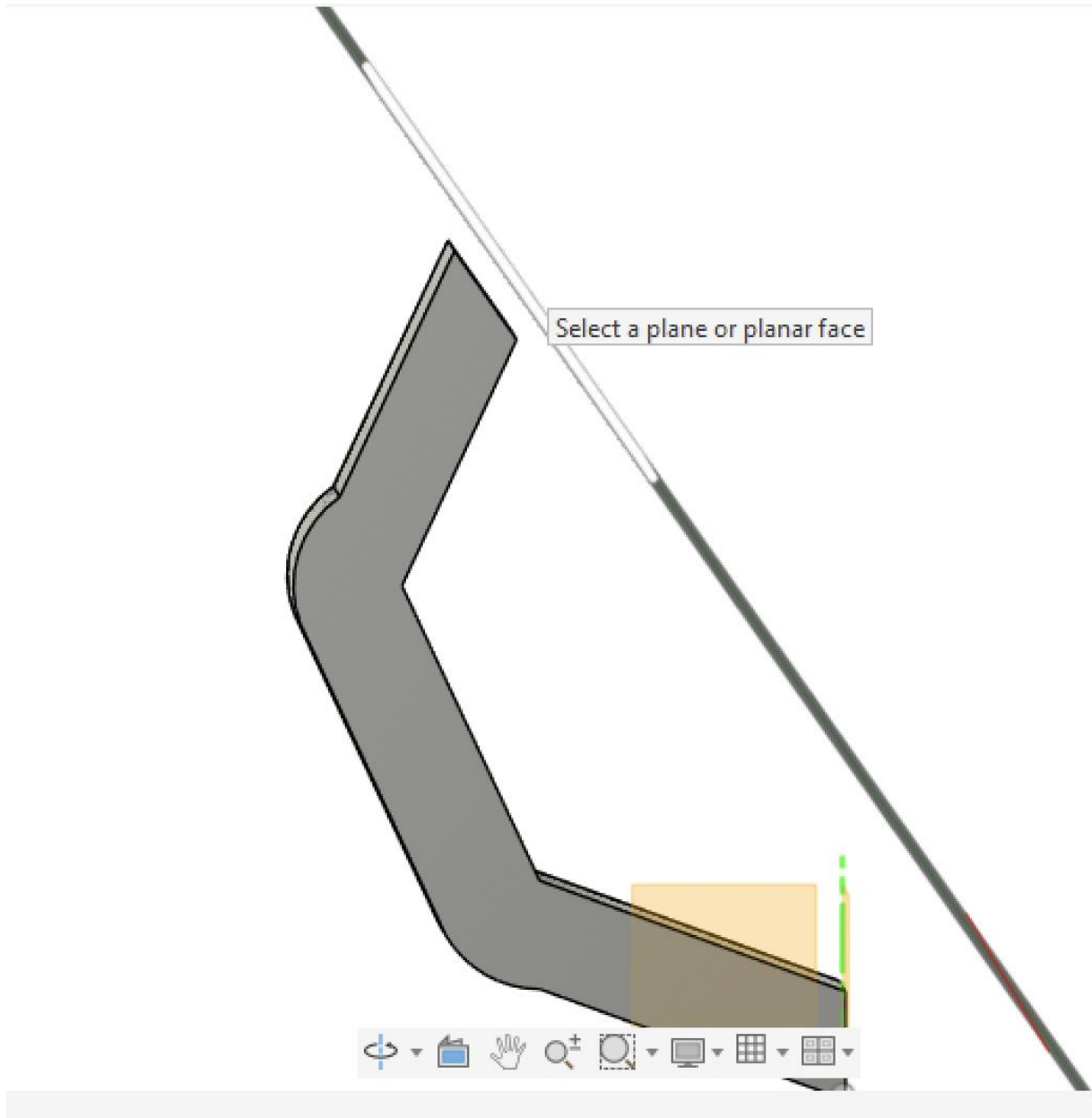


38. Create an offset plane 3mm from the face of the sweep created in the last step.
- a. The “Offset Plane” tool creates a **new plane parallel to an existing surface** and moves it a specified distance away. This gives you a new location to sketch without changing the existing geometry.
 - b. You can now create sketches on this new plane!

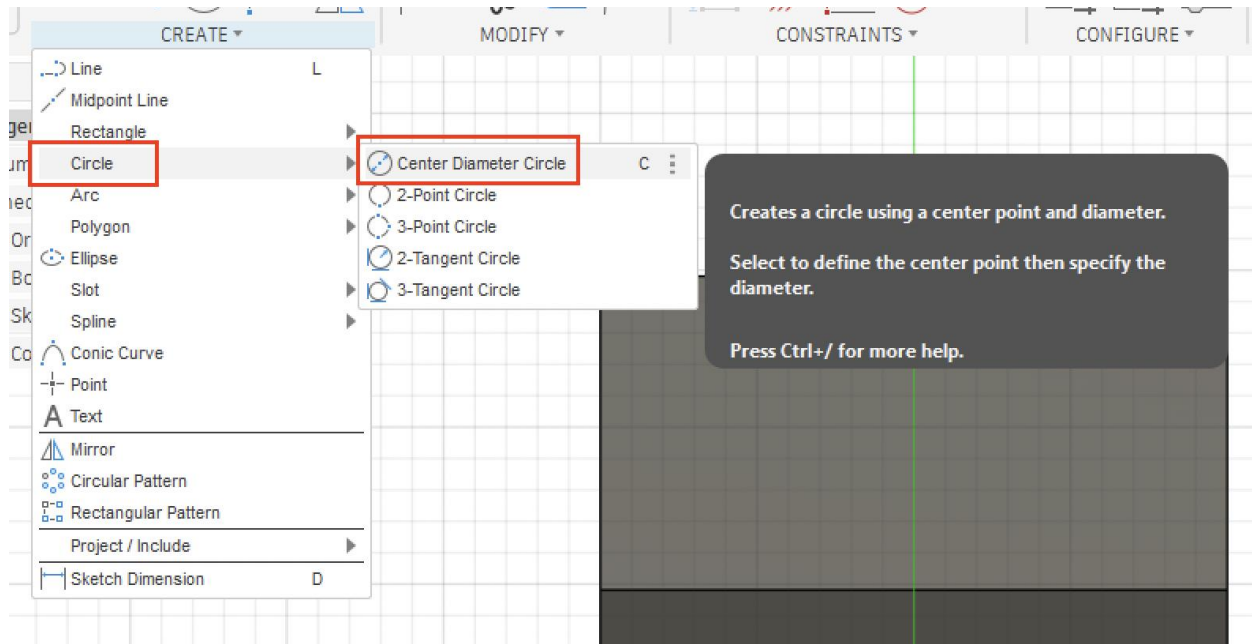


39. Create a sketch on the new offset plane.

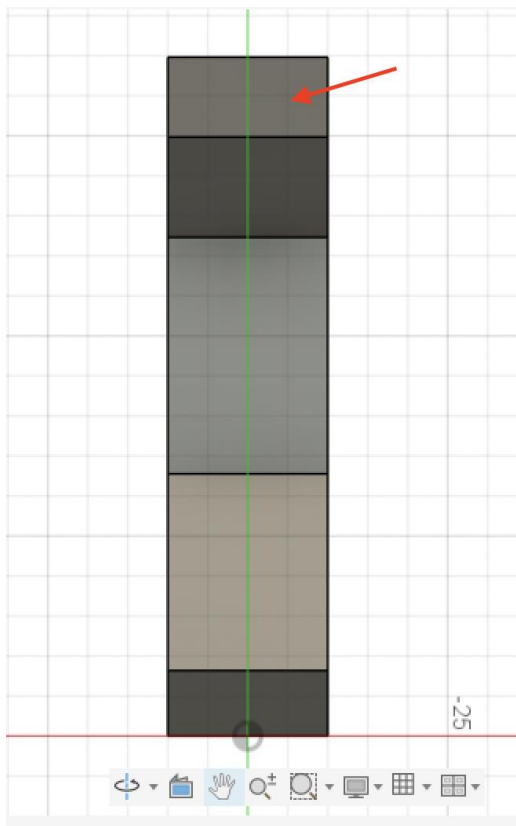
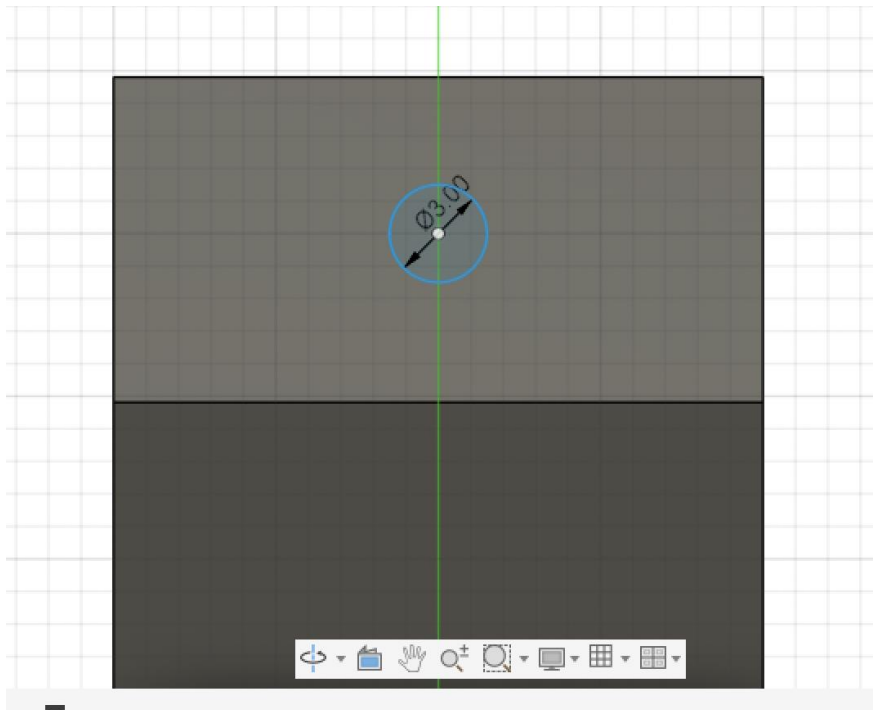
- a. Ensure your sketch is made on the plane itself, rather than the finger. Reference the image below.



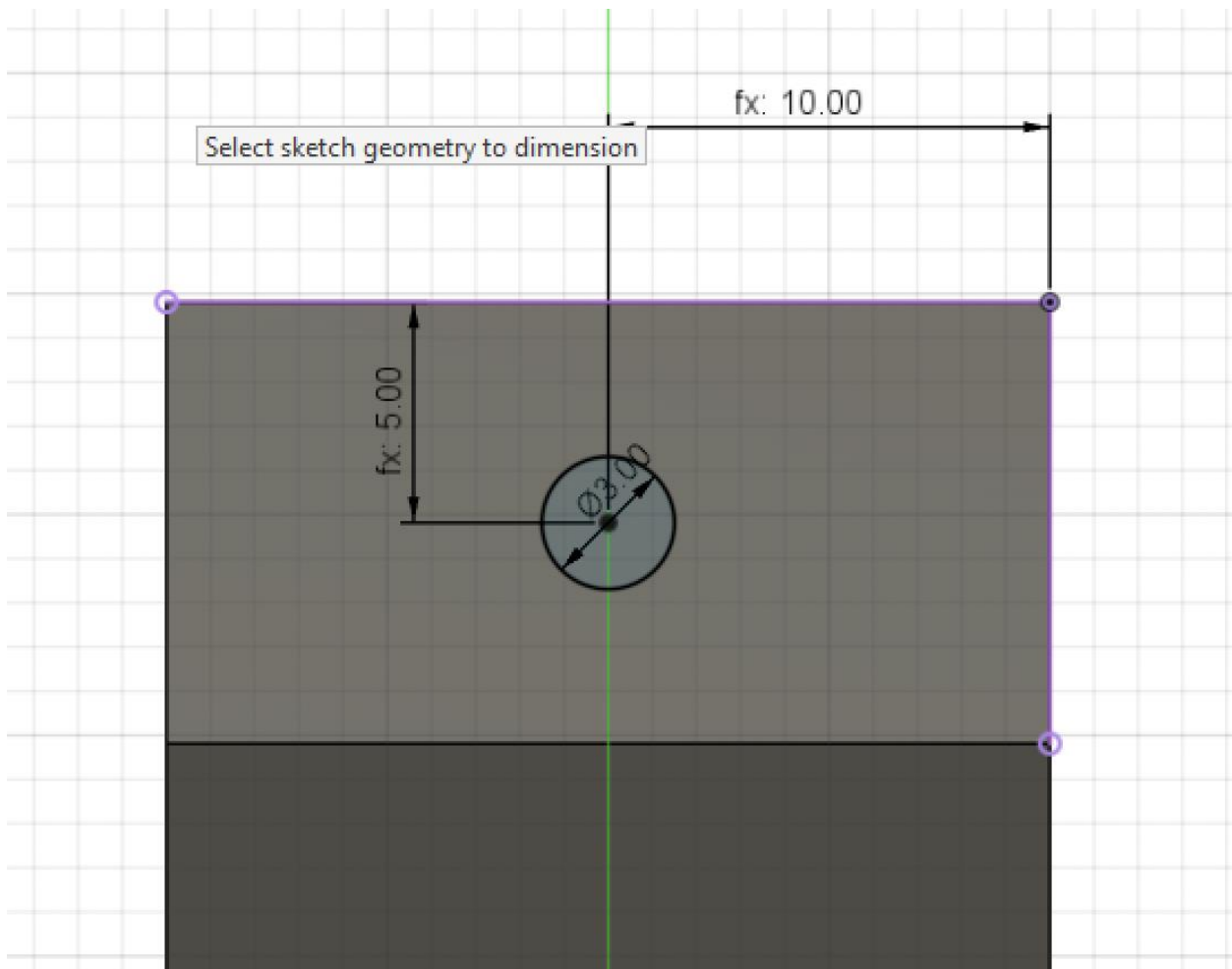
40. Navigate to the **“Create”** menu at the top of your screen and find the **“Circle”** option. A side menu should pop up. In this menu, choose **“Center Diameter Circle”**.



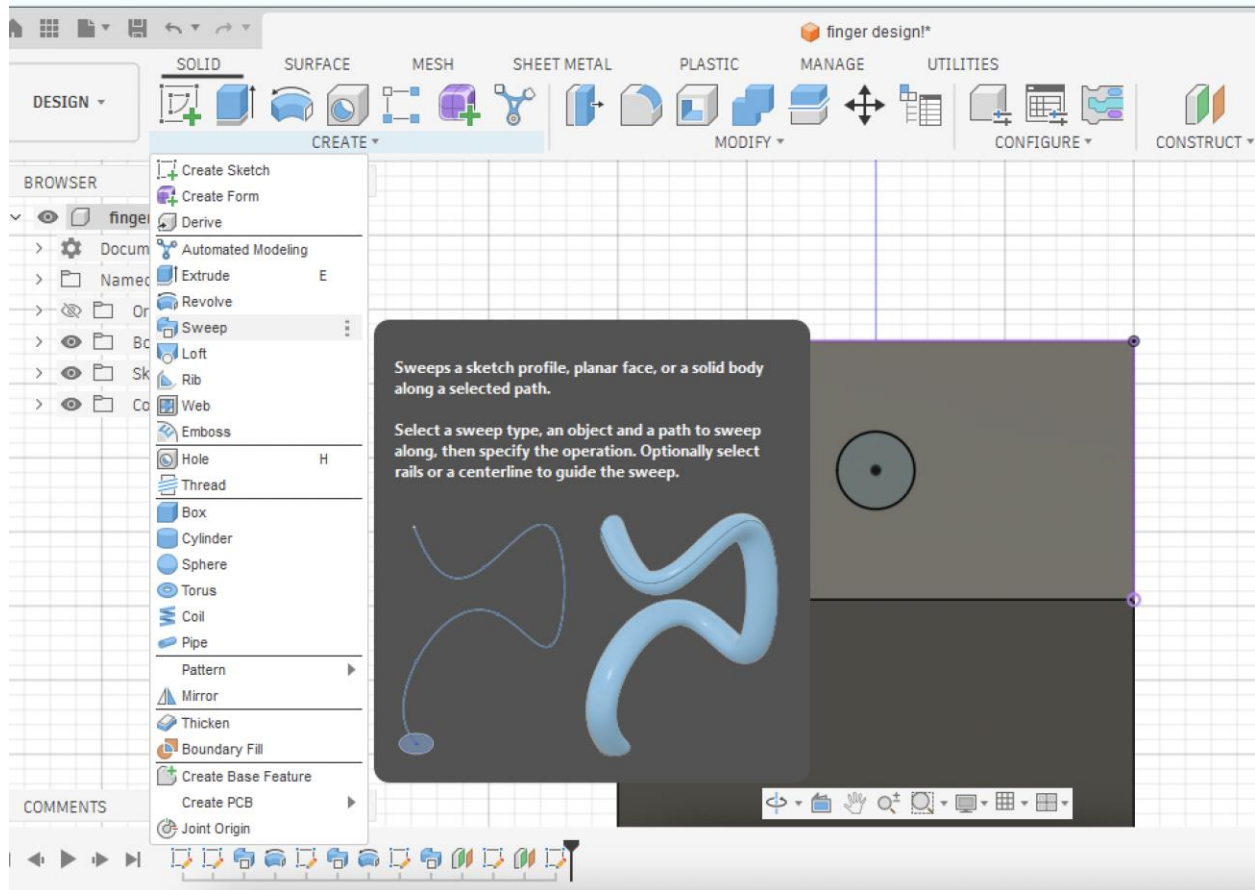
41. Create a circle in the middle of the top rectangle (rectangular face of the sweep we created in step 35). The diameter will be 3 mm.



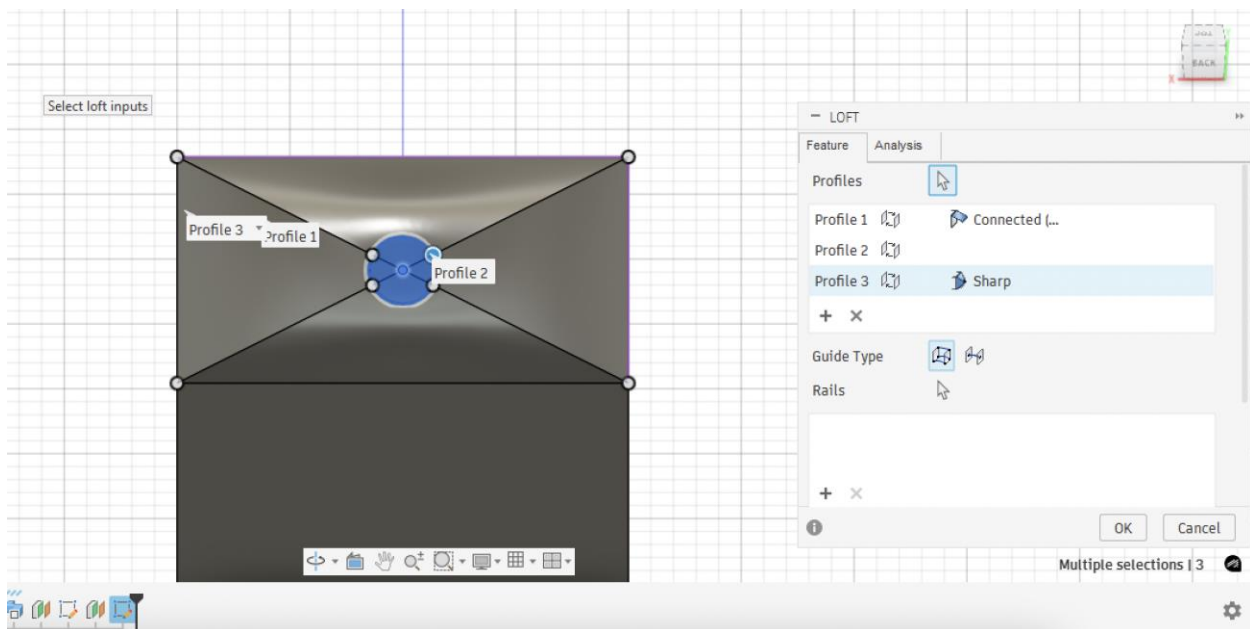
42. To center this circle in the middle of the rectangular face, use the “**Sketch Dimension**” tool to fix the lengths from the center of the circle to both the top and side of the rectangle.
- Use the **finger_width** variable for the **10.00 measurement fix** (distance from center of circle to side of rectangle).
 - Use the **finger_height** variable for the **5.00 measurement fix** (distance from the center of the circle to the top of the rectangle).
 - Hint:** Multiply the variable by a certain number to achieve the desired value. You used this method in step 34!



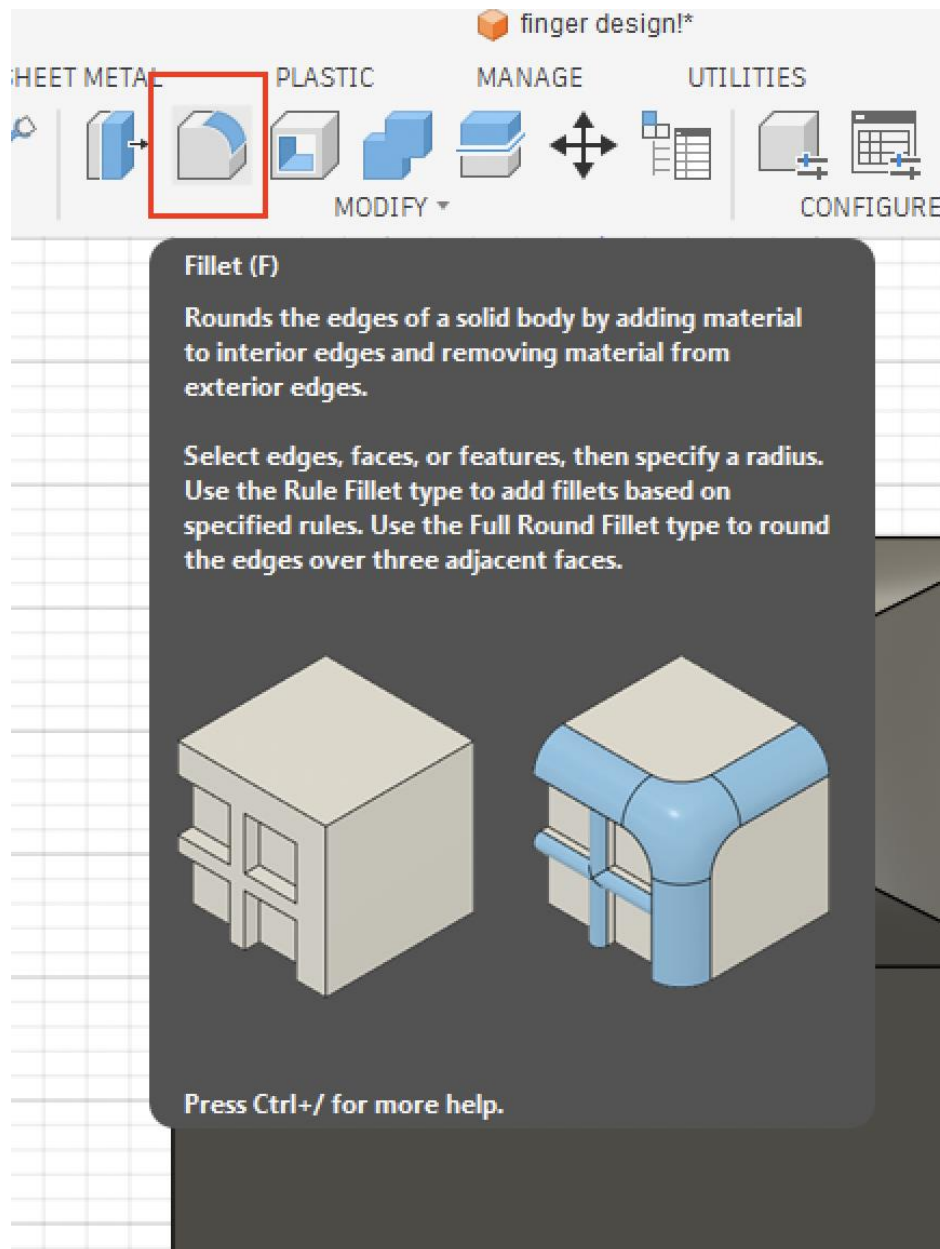
43. Once you **“Finish Sketch”**, navigate to the **“Create”** menu and find the **“Loft”** tool.



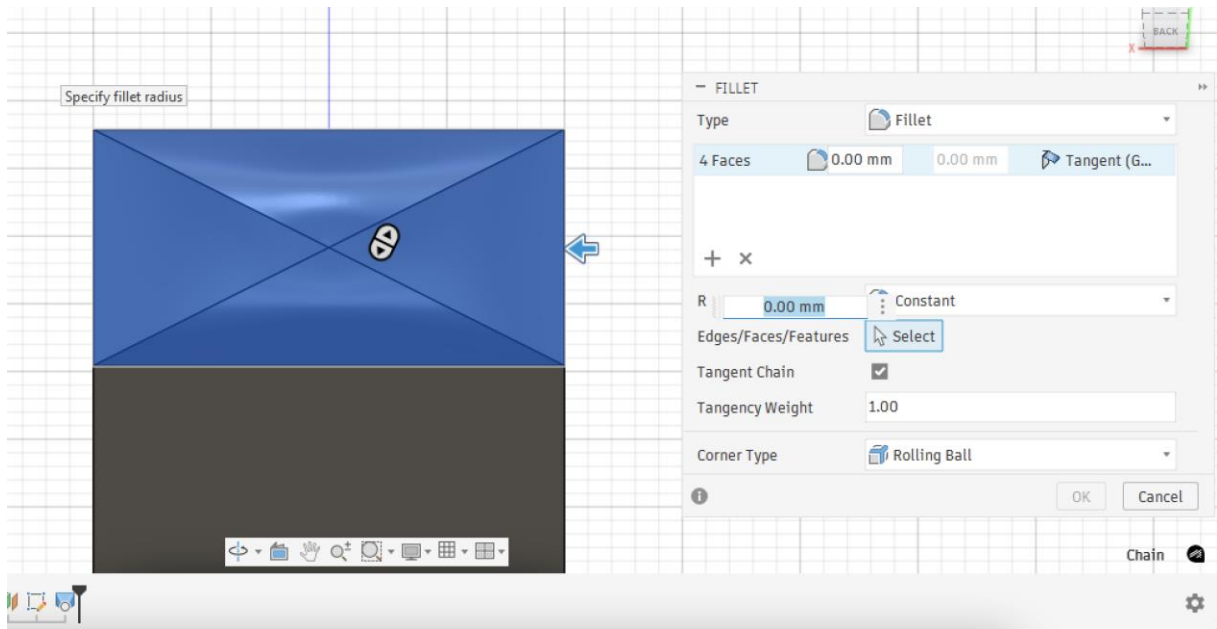
44. Once you select “**Loft**”, you can pick profiles. Select the rectangular plane you just sketched your circle on (Profile 1), the outside of the circle (Profile 2), and the center of the circle (Profile 3). Press “OK” after you finish.
- The **Loft** tool creates a smooth 3D shape by **connecting two or more profiles**. Fusion gradually transitions the geometry from the shape of one profile to the shape of the next.
 - In this step, Loft connects the **end face of the finger** to the **smaller profile on the offset plane**, creating the fingertip!



45. Navigate to the “Fillet” tool icon above the “Modify” menu and select.

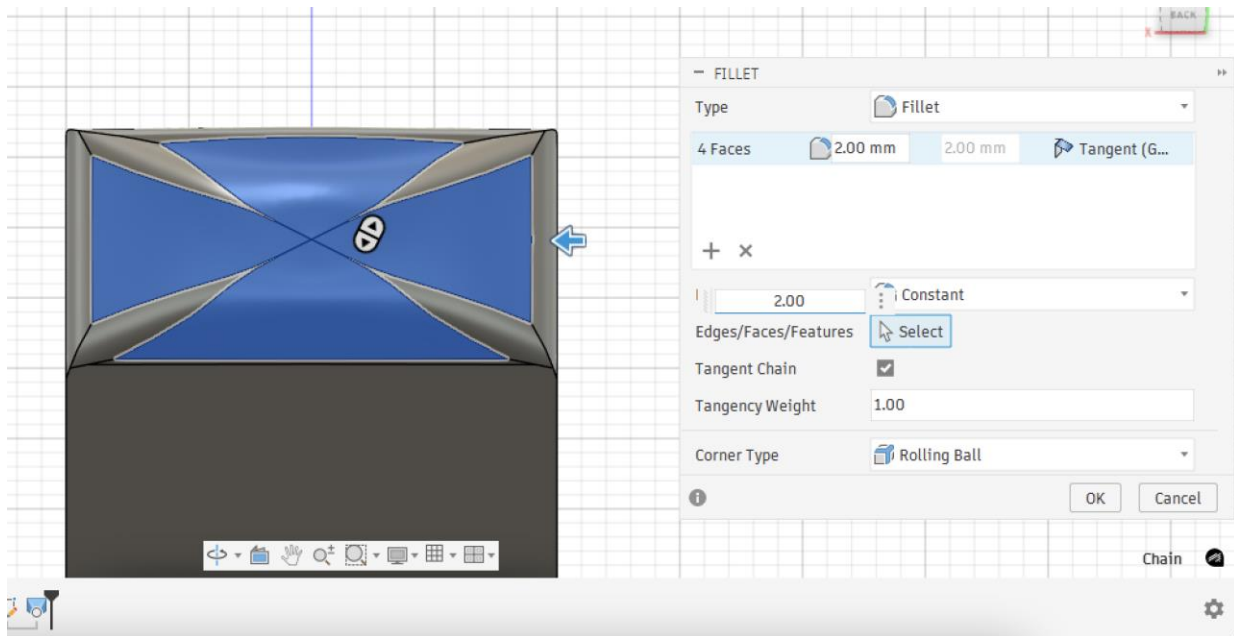


46. Click on the middle of the plane to select all faces.

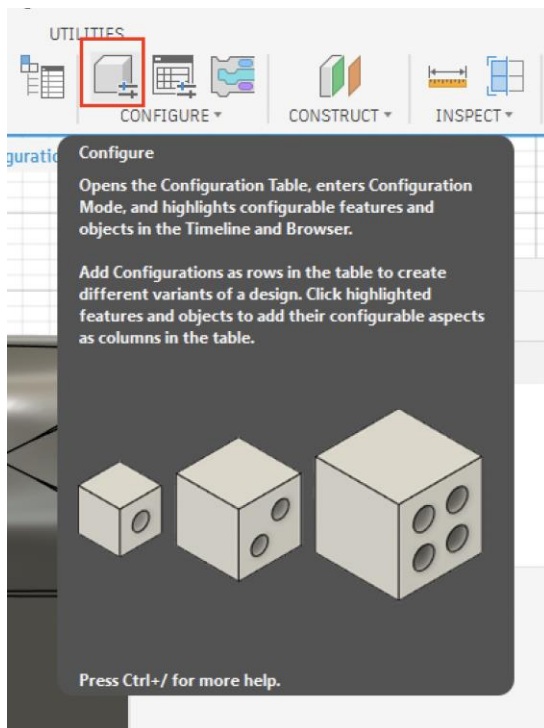


47. Change the radius to 2 mm. Then press “OK”.

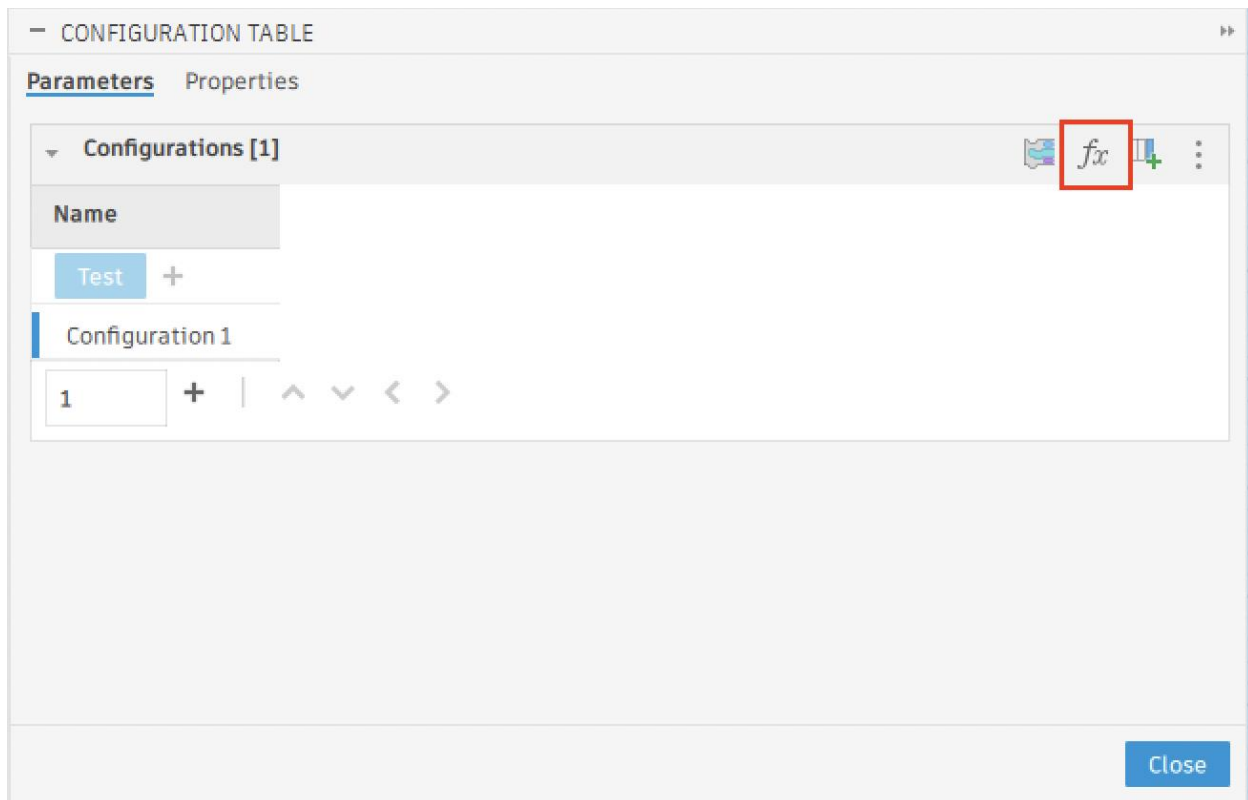
- a. The **Fillet** tool rounds selected edges by replacing sharp corners with a **smooth, curved transition**. The fillet radius determines how rounded the edge becomes. In this step, Fillet rounds the edges of the newly created fingertip, creating a **smoother and more natural transition** between the fingertip surfaces.



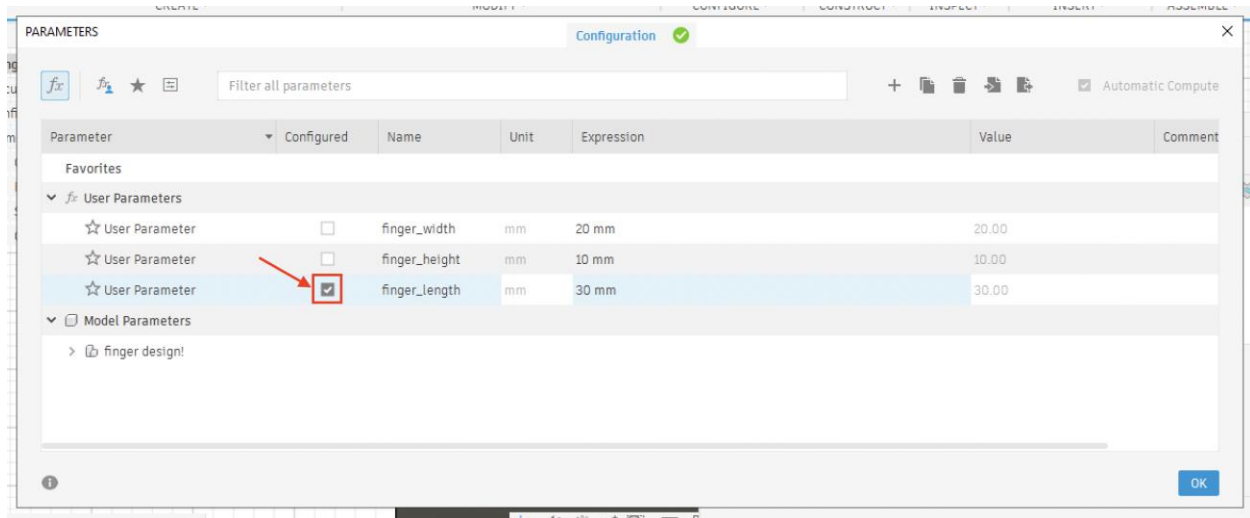
48. Navigate to the **“Configure”** tool icon above the **“Configure”** menu at the top of your screen.



49. When the configuration table pops up, navigate to the fx icon in the top right and click on it.
- Configurations** allow you to create multiple versions of the same model by changing selected parameters or features.



50. When your parameters table pops up, click on the **checkmark box** next to the finger_length variable to select it. Press “OK” after this is complete.



51. Use the **add button** at the bottom of the chart to add 5 total configurations. Add the configurations shown in the image below (. Click “Close” after you finish.

- Instead of creating a separate CAD model for every finger size, each configuration stores a different value for finger_length within the **same finger model**.
- Adding finger_length to the configuration table allows each configuration to have its **own value for finger length**. When you switch between configurations, Fusion changes finger_length to the value listed in the table and automatically updates the finger geometry.

CONFIGURATION TABLE

Parameters

Properties

Configurations [1]

Name	finger_le...
Test +	30 mm
Configuration 1	30 mm

1 + ^ v < >

Add Configurations

Adds new Configurations to the Configured Design with column values that match the active Configuration.

Close

CONFIGURATION TABLE

Parameters

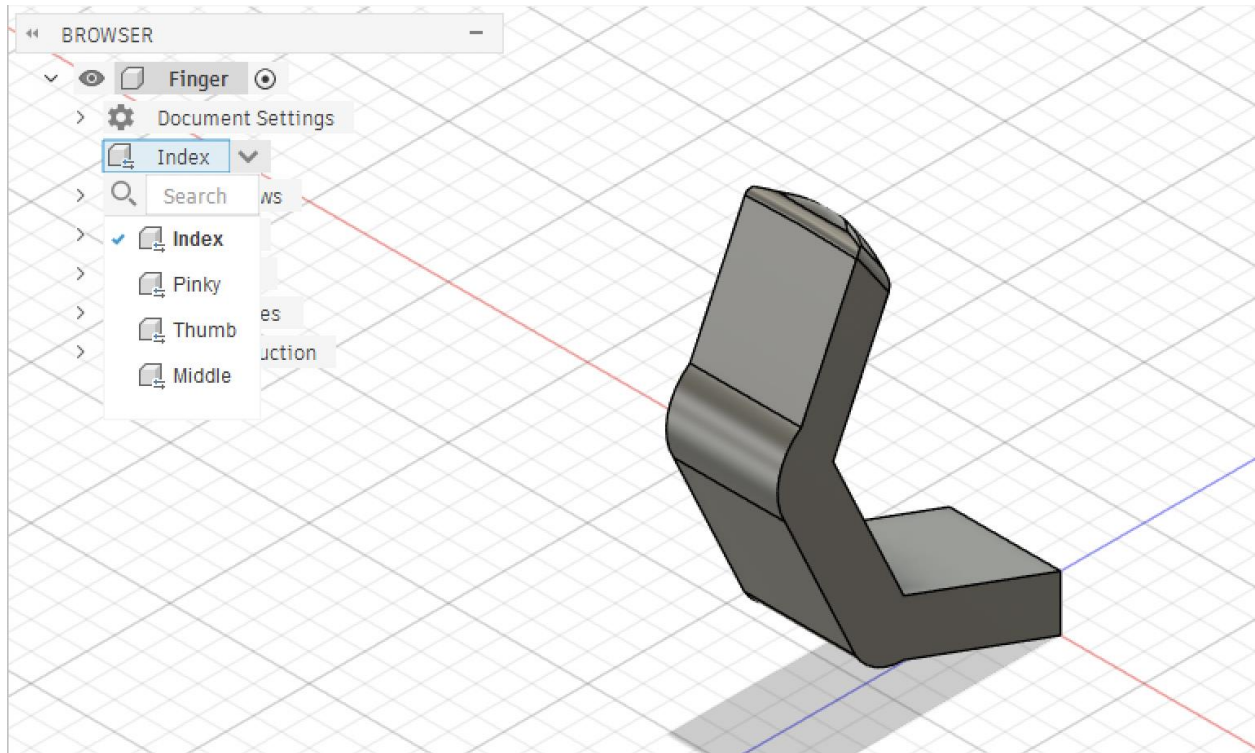
Properties

Configurations [5]

Name	finger_len...
Test +	32 mm
Index	30 mm
Pinky	25 mm
Thumb	15 mm
Middle	32 mm
Ring	30 mm

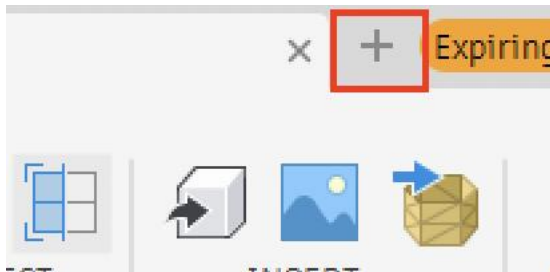
1 + ^ v < >

52. You can shift through the different finger configurations by opening the menu under “Document Settings”. [Check out how each figure looks based on the parameters you inputted!](#)

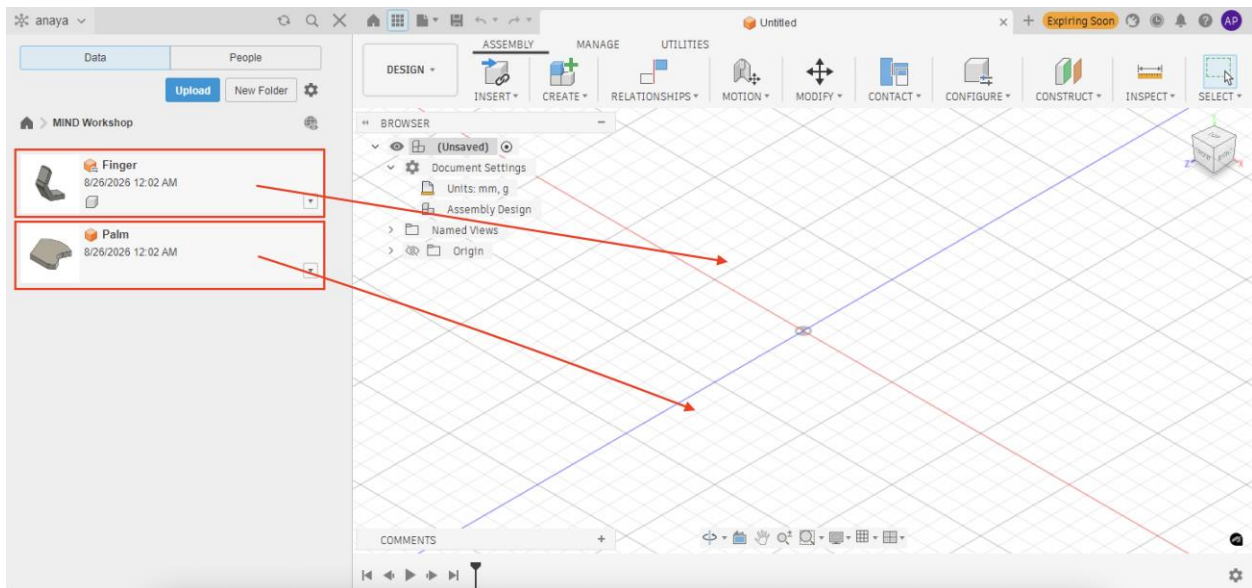


Part 4: *Hand Assembly*

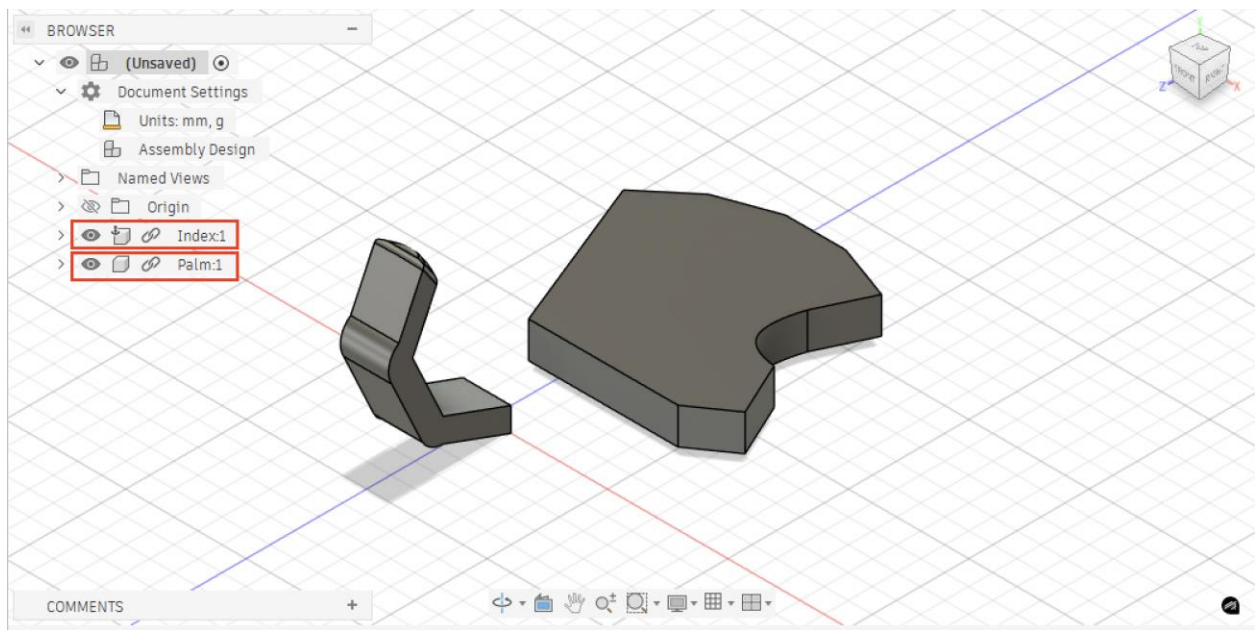
53. Create a new assembly design by adding a tab in Fusion, near the top of the screen, and selecting “**Assembly Design**”.



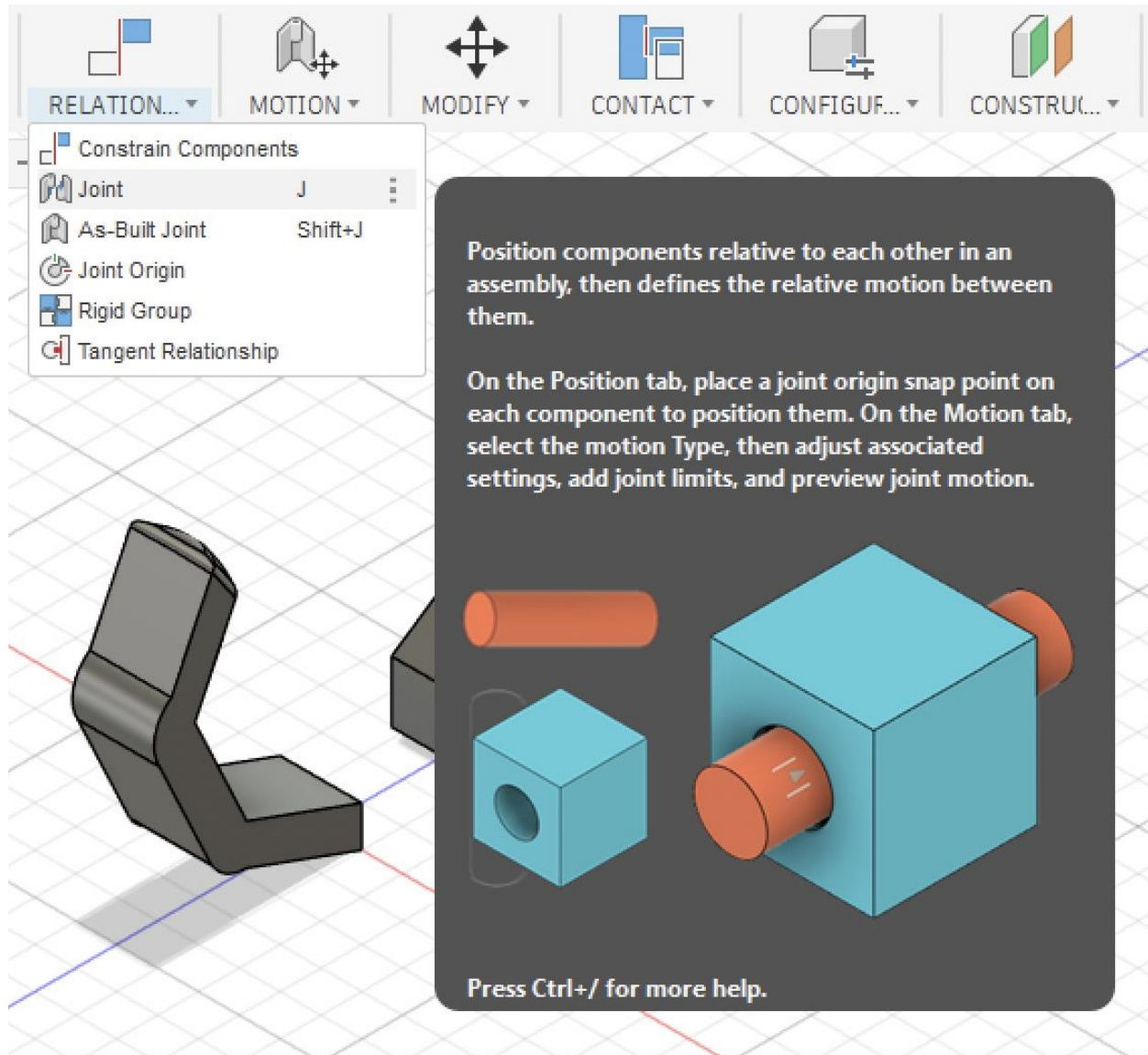
54. Open your data panel (3x3 dot matrix icon) and open up your “MIND Workshop” project.
55. **Drag and drop** your “Palm” and “Finger” design into your blank assembly design workspace.
- When dropping your finger in, you will be asked to select your desired configuration. Pick any!



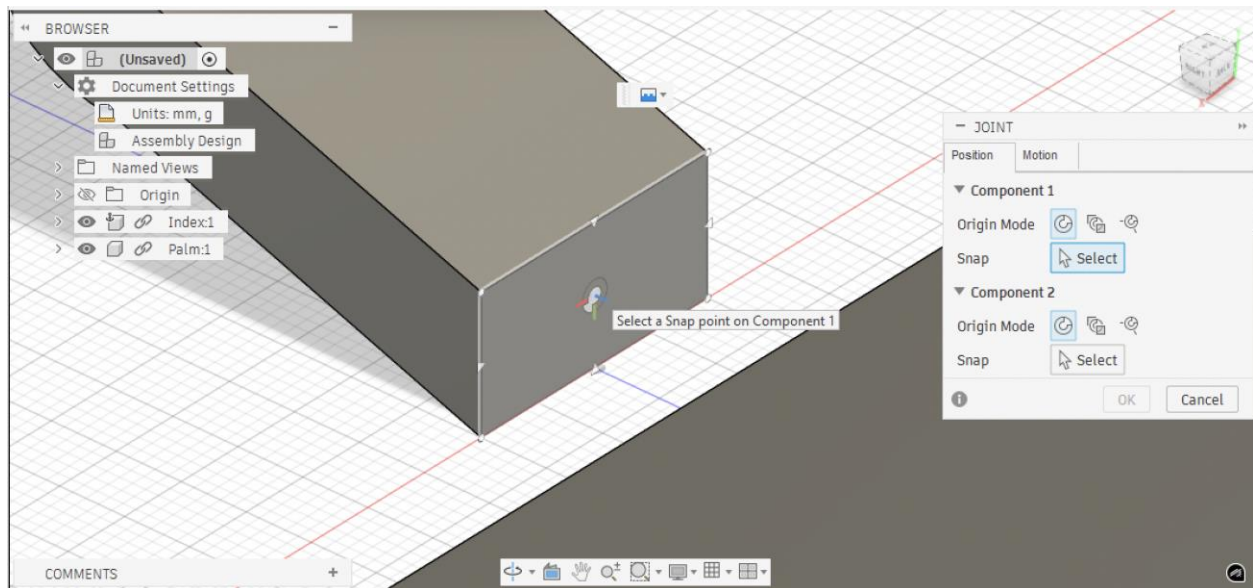
You can click on either of your designs on the left menu to select the design. Once the design is selected, you can move it around the page.



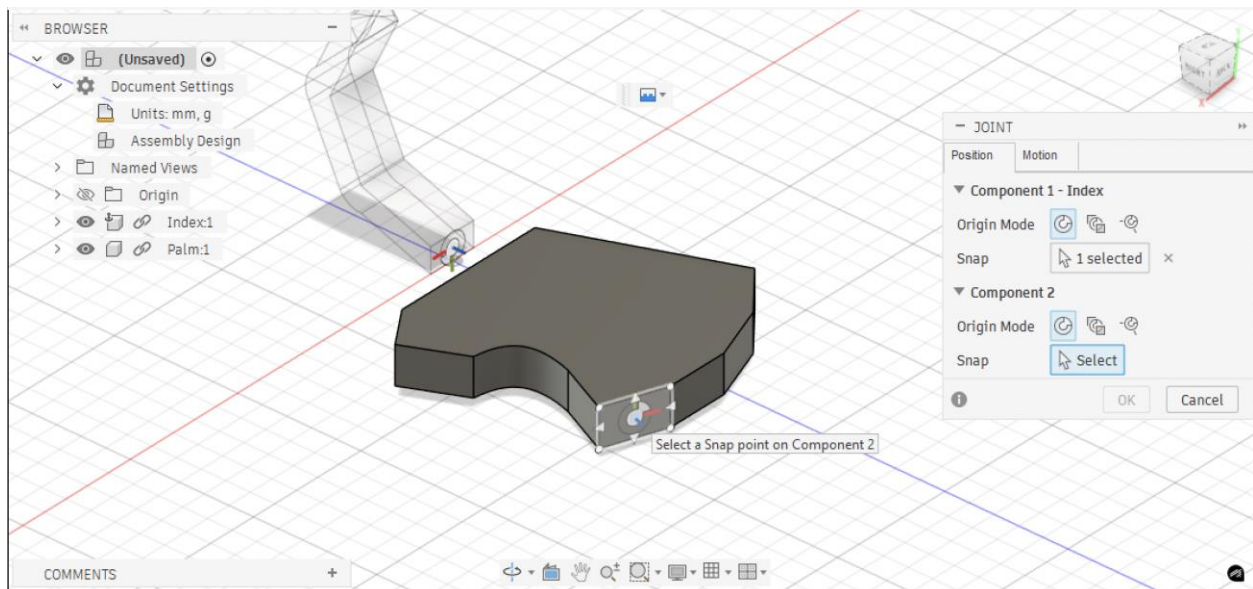
56. Navigate to the “**Relationships**” menu and click on the “**Joint**” option.

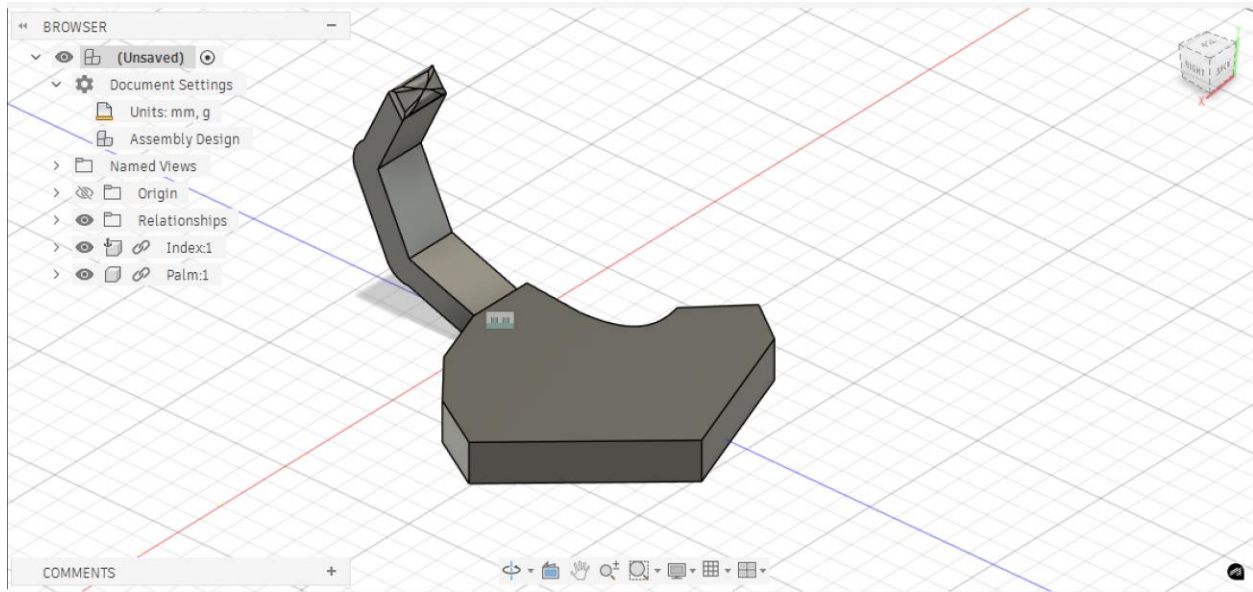


57. Click the center of the rectangle you created in the first sketch of your finger design as the “**Snap**” point of your “**Component 1**”.

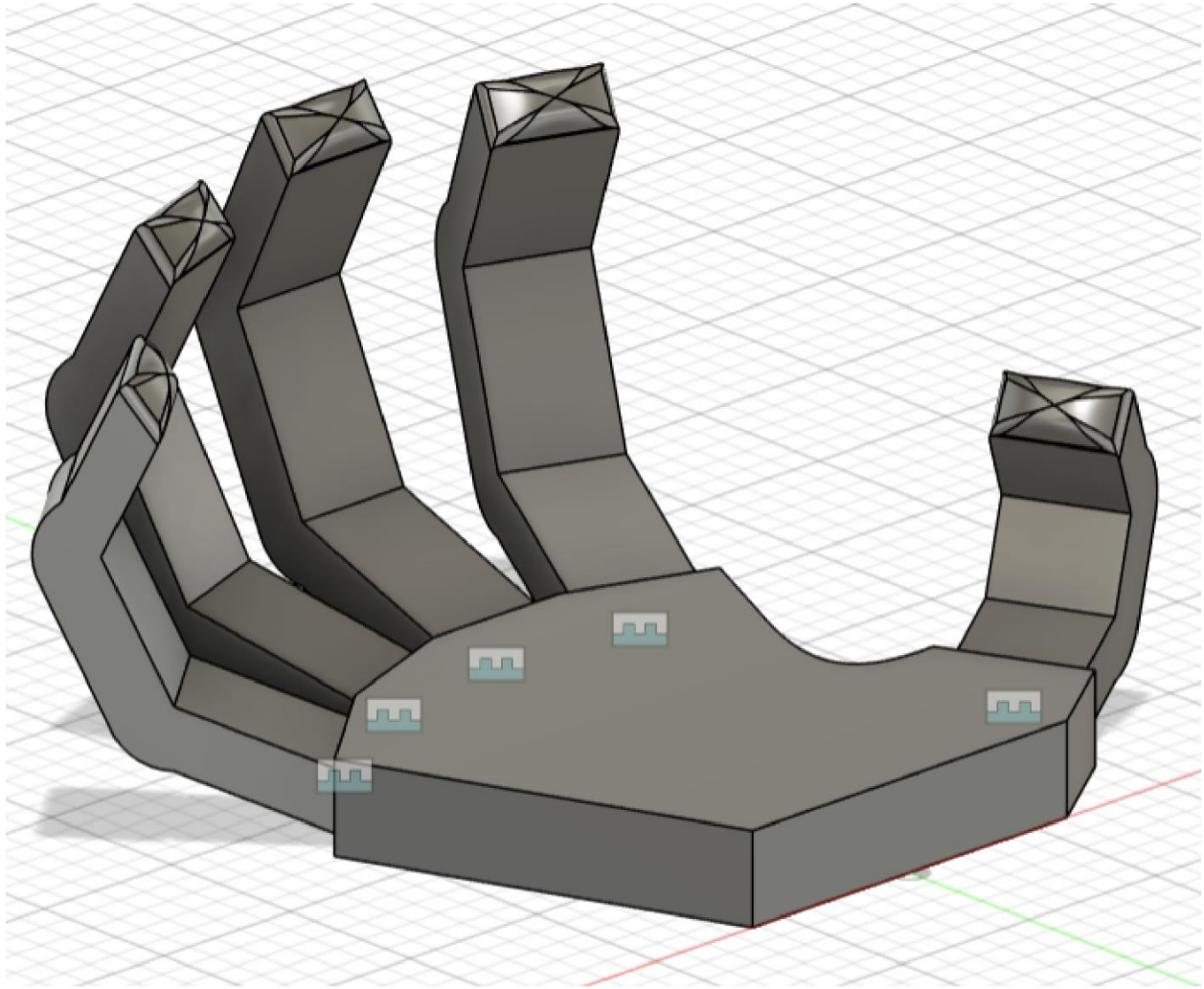


58. Click the center of the rectangle where your chosen finger would be connected to the palm as the “**Snap**” point of your “**Component 2**”.
- You can keep the degree value as 0. (Should automatically be 0).
 - You now have one finger connected to your palm!





59. Drag and drop another finger and repeat these steps (steps 55-57) for every remaining finger based on the configurations you pick.



Congrats, you have completed your final model! Make sure to save your design (name it anything you want).

Find a member of MIND Leadership and get checked off!